GLM - Assumptions

Lecture 11 - LAS

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09-11-2026

Today

Recap
Assumptions
Residuals
Normality
Constant Error Variance
Independence
Conclusions

Trustworthy Faces

Recall The One Million Impressions (OMI) dataset, which contains over a million human judgments of over 1,000 synthetic face images along 34 attributes.

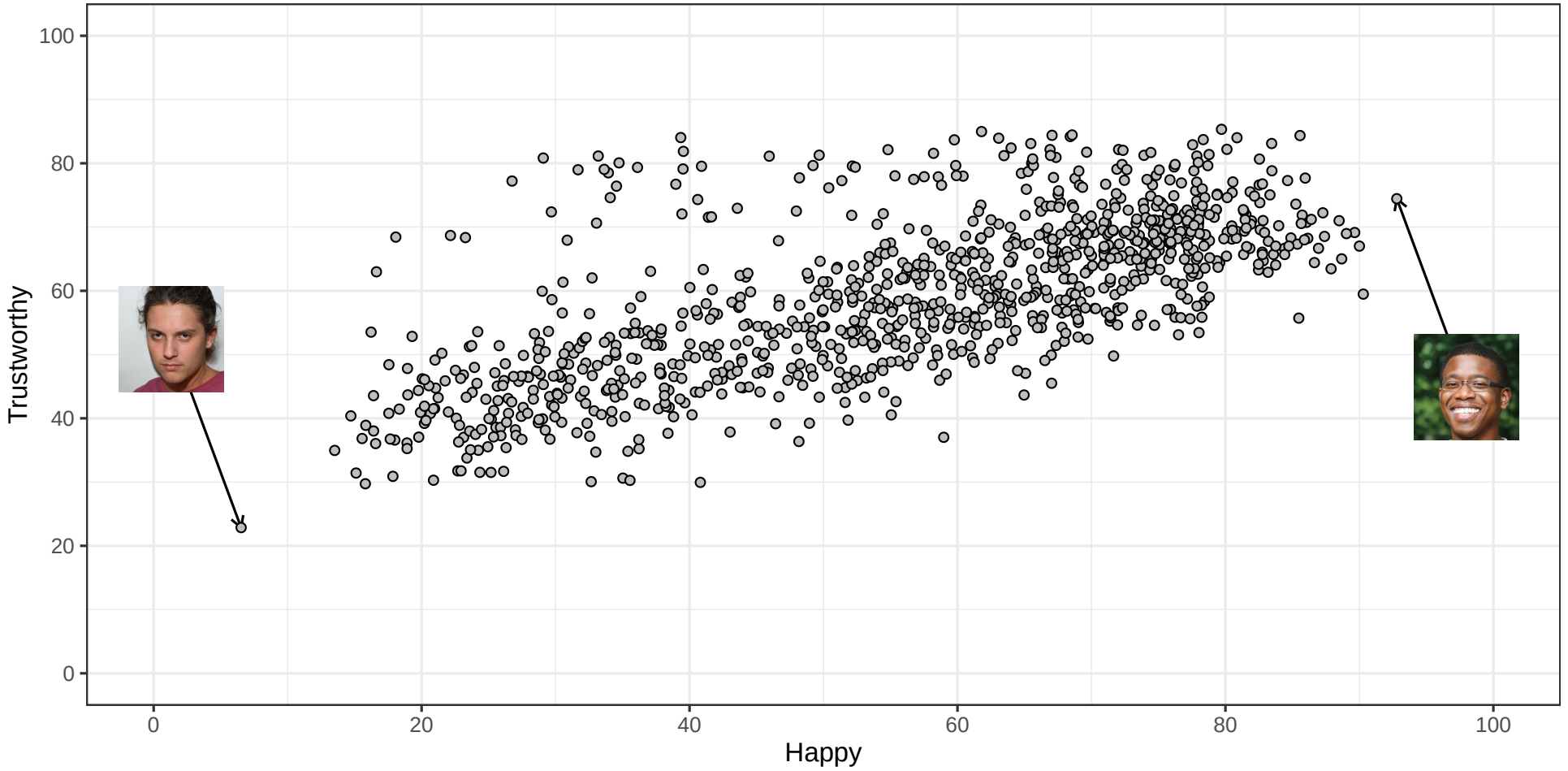
Participants judged faces on a scale from 0 to 100 on attributes such as

- trustworthy
- attractive
- smart
- memorable
- dominant
- well-groomed

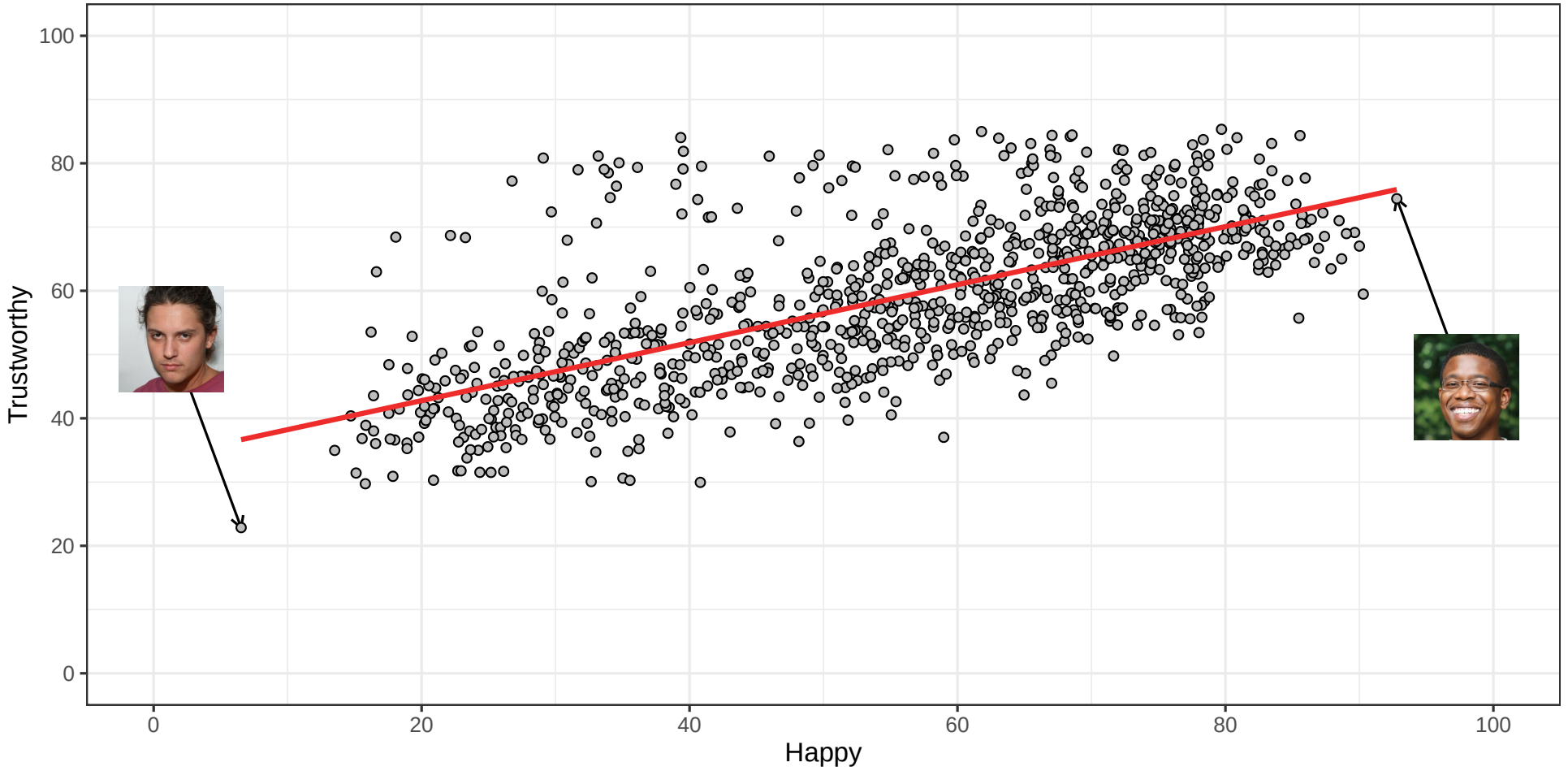
Each observation is the average rating across many participants.

Peterson, J. C., Uddenberg, S., Griffiths, T. L., Todorov, A., & Suchow, J. W. (2022). Deep models of superficial face judgments. Proceedings of the National Academy of Sciences (PNAS), 119(17), e2115228119. doi:10.1073/pnas.2115228119 source: https://github.com/jcpeterson/omi/

Trustworthiness vs. Happiness



Trustworthiness vs. Happiness



Trustworthiness vs. Happiness

Trustworthiness vs. Happiness

Testing Coefficients

Compute

$$t = \frac{a - \alpha_{\mathcal{H}_0}}{\mathrm{SE}_a} \qquad \text{with } df = N - 2$$

- Examine the t-table
  - use the appropriate tail of the distribution depending on the hypothesis test conducted.

For the slope everything worked out the same, but with b instead of a.

Comparing two models

Rather than individually comparing a and b against 0 or specific values, we can also compare two models:

$$\begin{aligned} \mathcal{H}_0 &: \text{the null model}, & y_i &= a + 0 \times x_i + e_i \\ \mathcal{H}_1 &: \text{the regression model}, & y_i &= a + b \times x_i + e_i \end{aligned}$$

1. $\mathcal{H}_0 : \beta = 0$ vs $\mathcal{H}_1 : \beta \neq 0$.

2. $\mathcal{H}_0 : \rho^2 = 0$ vs $\mathcal{H}_1 : \rho^2 > 0$.

Here, ρ^2 is the *population* value for R^2

Comparing two models

Mean of y

6.1

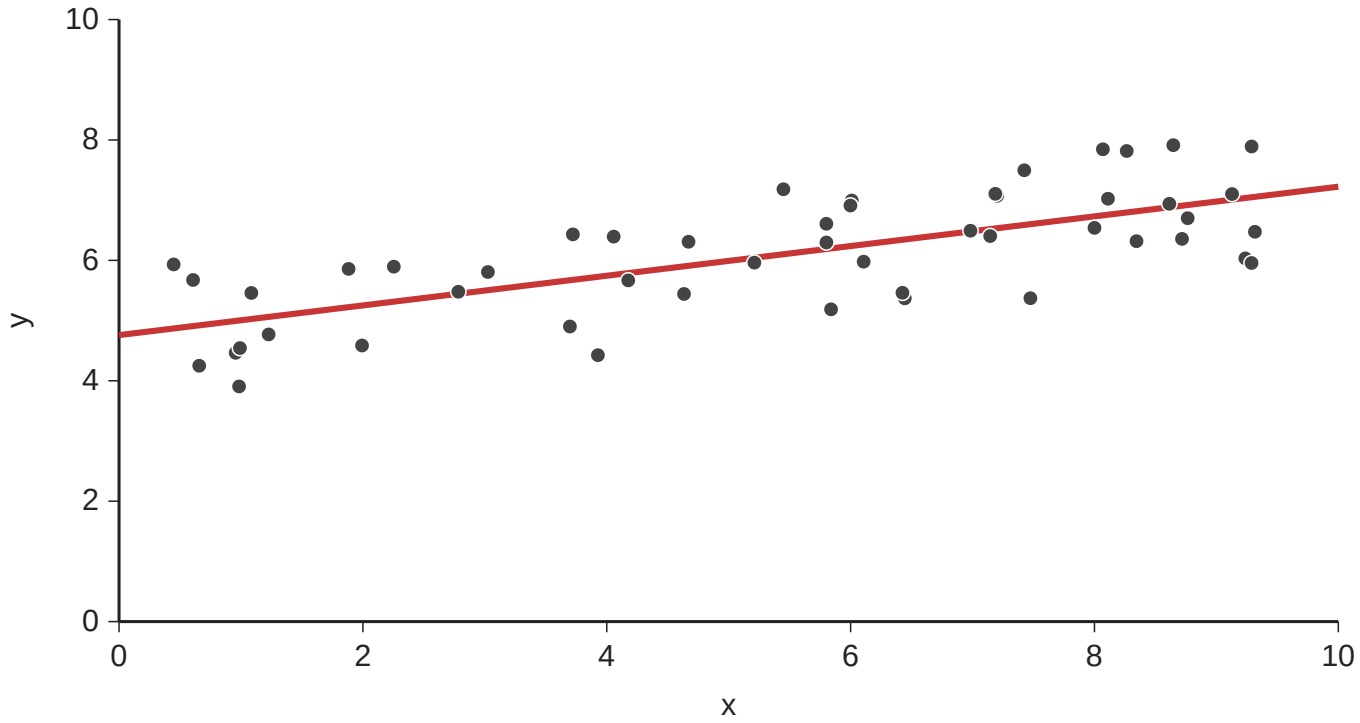
Variance of y

1.0

R^2

0.50

TSS	**49.00**
SSE	**24.50**
RSS	**24.50**



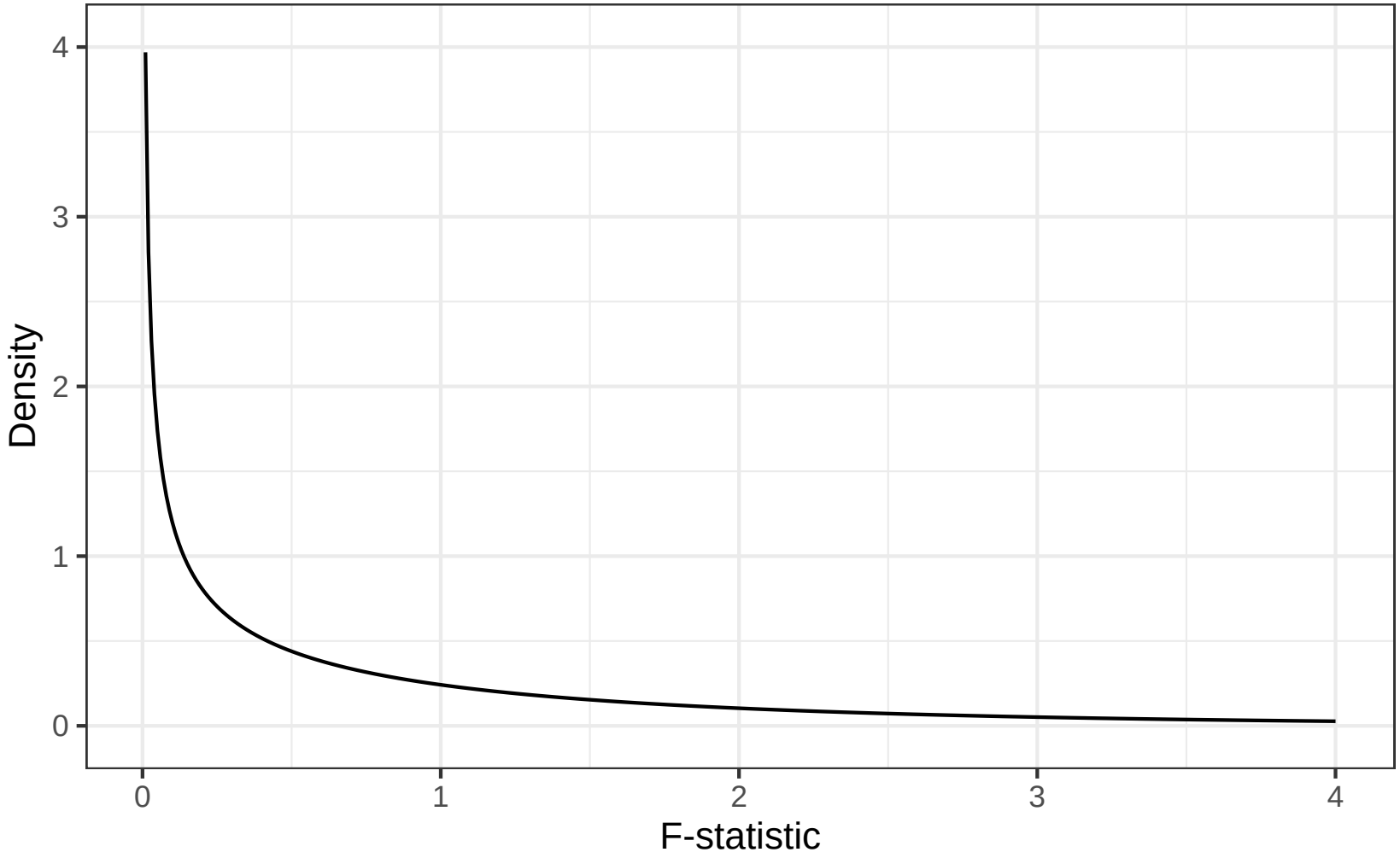
F-test

$$F = (N-2) \times \frac{R^2}{1-R^2} = (N-2) \times \frac{\text{Explained variance}}{\text{Unexplained variance}}$$

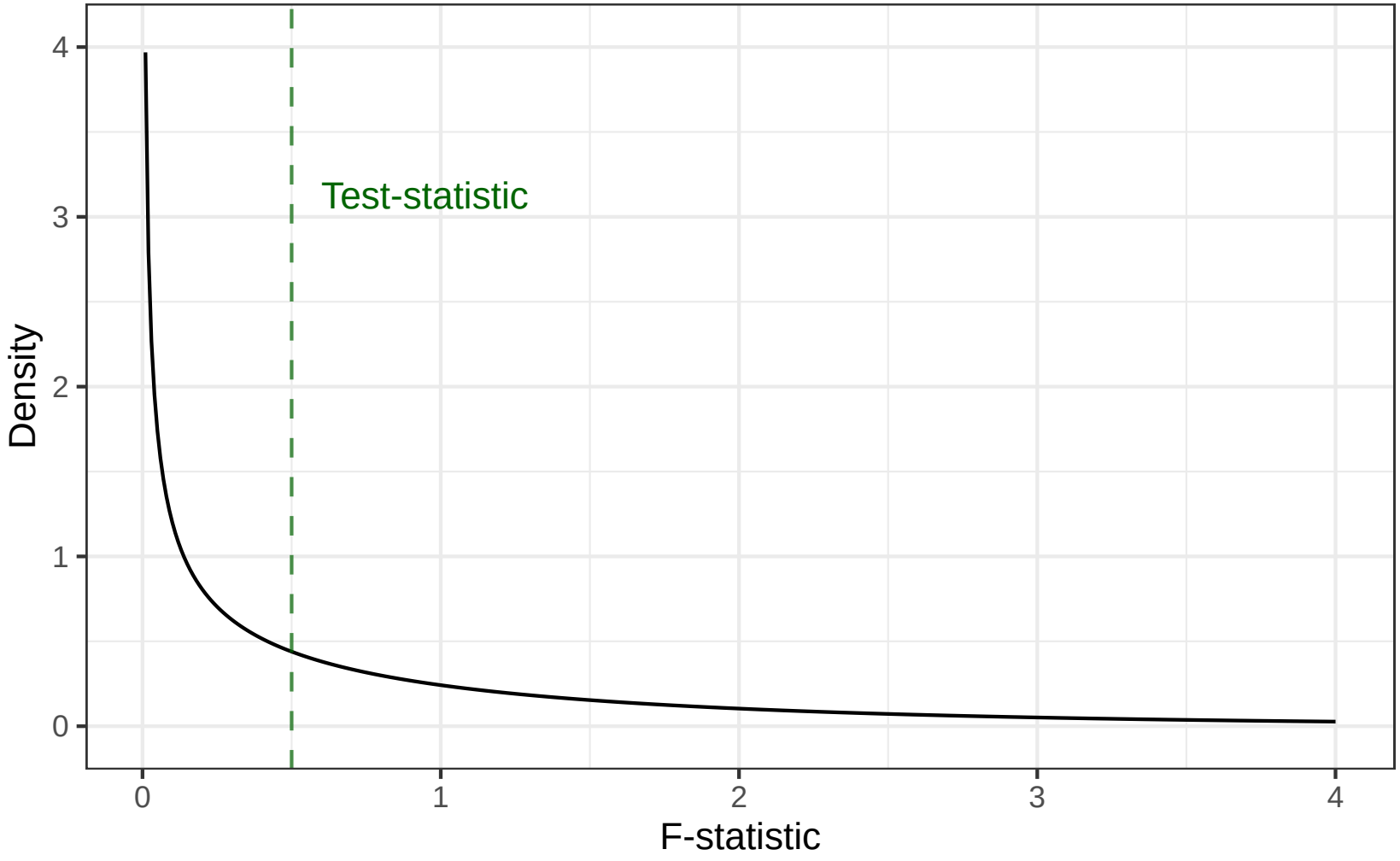
If $\mathcal{H}_0$ is true, the F-value is F-distributed:

$$F \sim F(1, N-2)$$

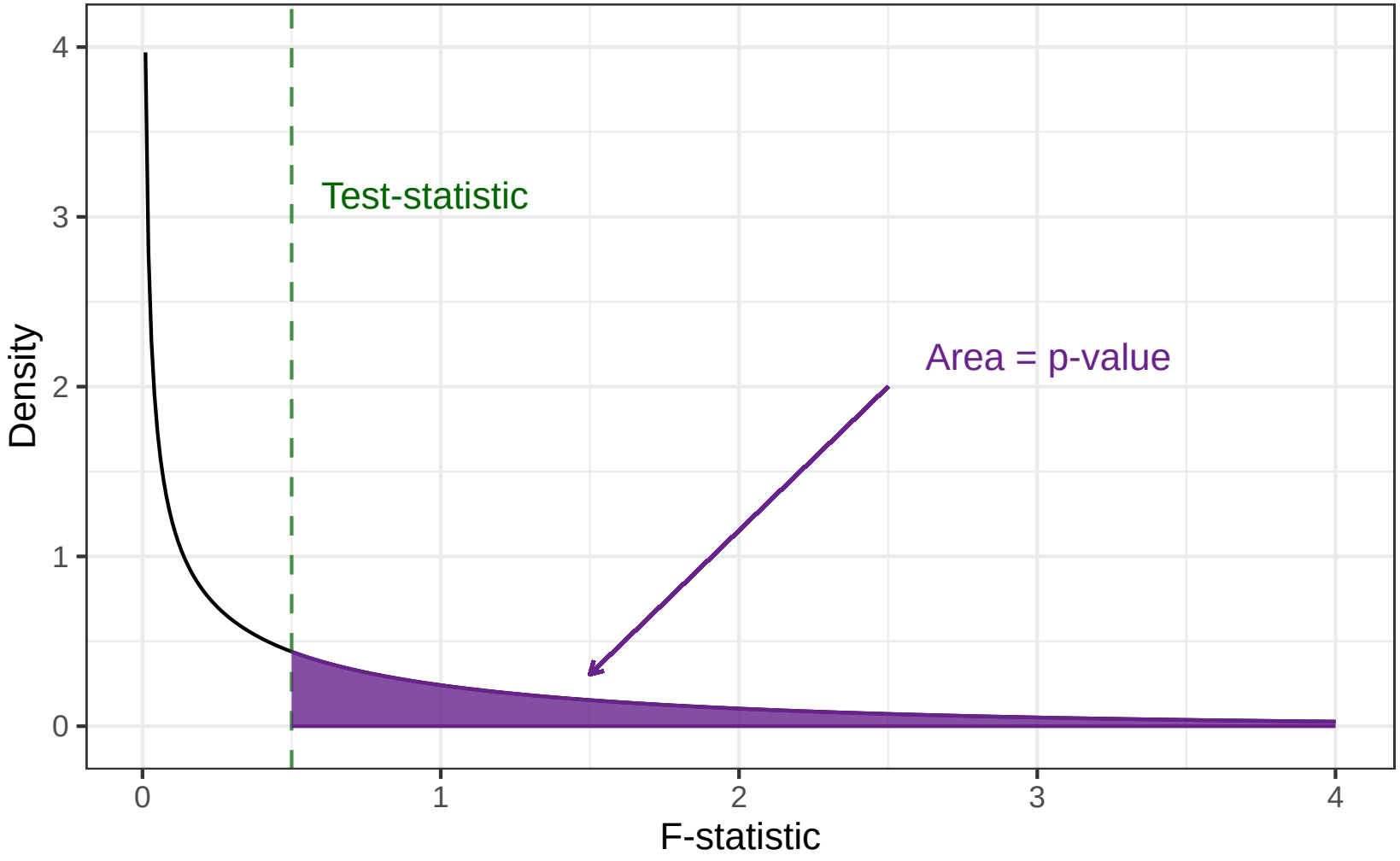
F-distribution



F-distribution



F-distribution



F-test: happy predicting trustworthy

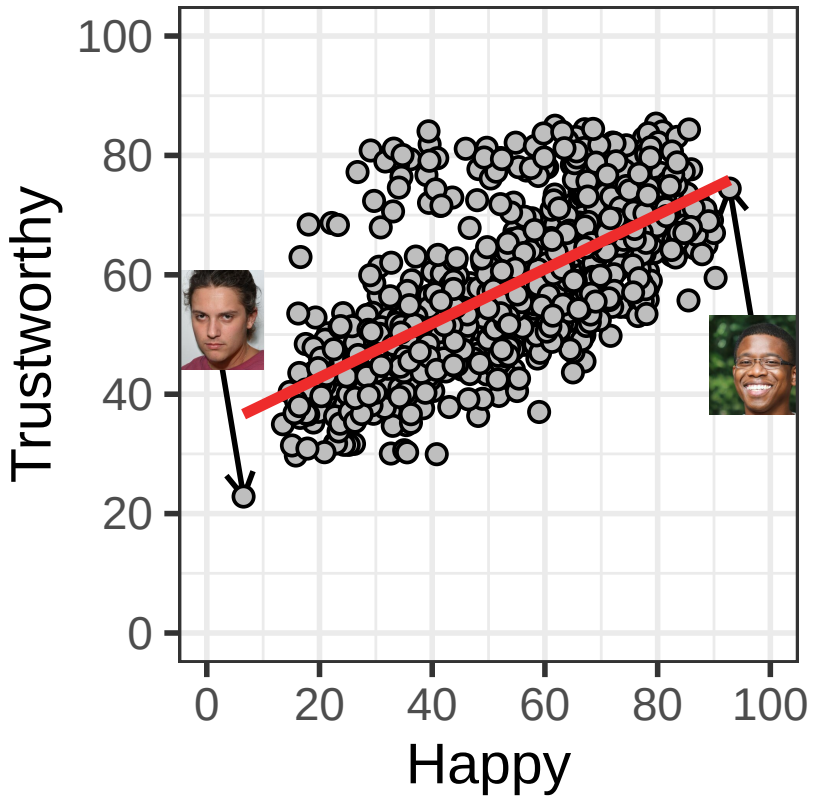
$N = 1,004$, $R^2 = 0.49$.

$$\mathcal{H}_0 : \rho^2 = 0, \qquad \mathcal{H}_1 : \rho^2 > 0$$

$$F = (N - 2) \times \frac{R^2}{1 - R^2} = 955.86$$

$F(1, 1002) = 955.86$, $p < .001$.

We reject $\mathcal{H}_0$. Happy ratings explain about 49% of the sample variation in trustworthiness.



F-test: memorable predicting trustworthy

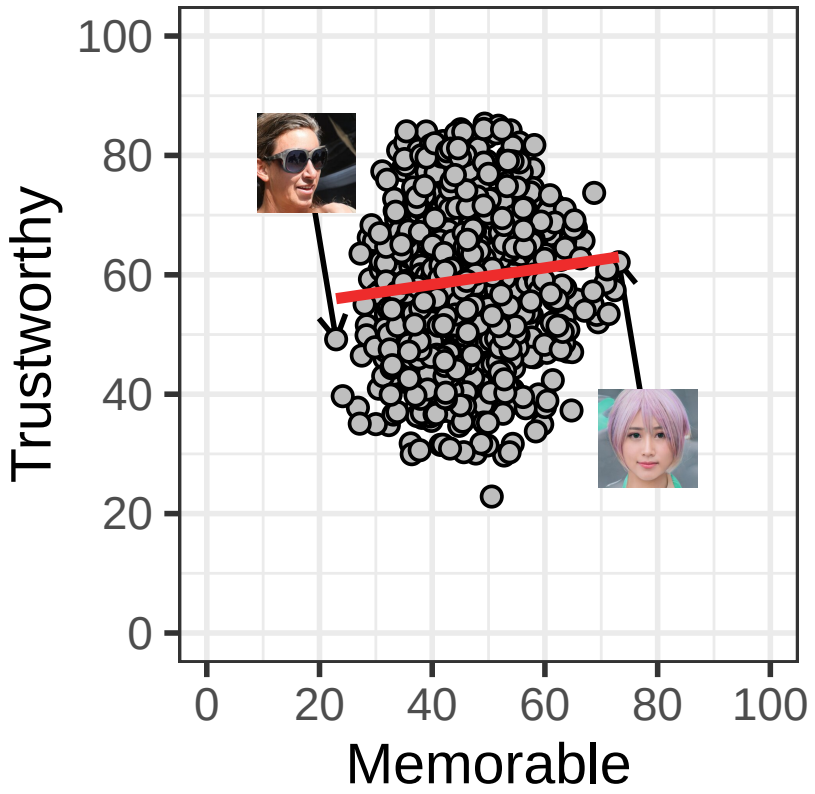
$N = 1,004$, $R^2 = 0.01$.

$$\mathcal{H}_0 : \rho^2 = 0, \qquad \mathcal{H}_1 : \rho^2 > 0$$

$$F = (N-2) \times \frac{R^2}{1-R^2} = 9.50$$

$F(1, 1002) = 9.50$, $p < .01$.

Statistically significant, but practically of limited value: Memorableness ratings explain only 1% of the sample variation in trustworthiness.



Recap

1. The Ordinary Least Squares finds a and b that minimize the SSE.
2. But SSE on its own doesn't tell us how good the predictions are.
3. R^2 tells us how much variance is explained by the regression line.
4. We test a and b using t-tests and compare models using the F-test.

Today:

- Linear regression, and these significance tests make assumptions.
- What are these assumptions?
- How do check if they are violated?

Today

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Assumptions

- Almost all models make assumptions. Assume that…
  - the data are normally distributed
  - the sample is representative for the population
  - observations are independent of each other

- If assumptions are violated things can go wrong:
  - The model could misrepresents the data
  - Tests are misleading
    - The type-I error is inflated
  - Inferences may not be justified

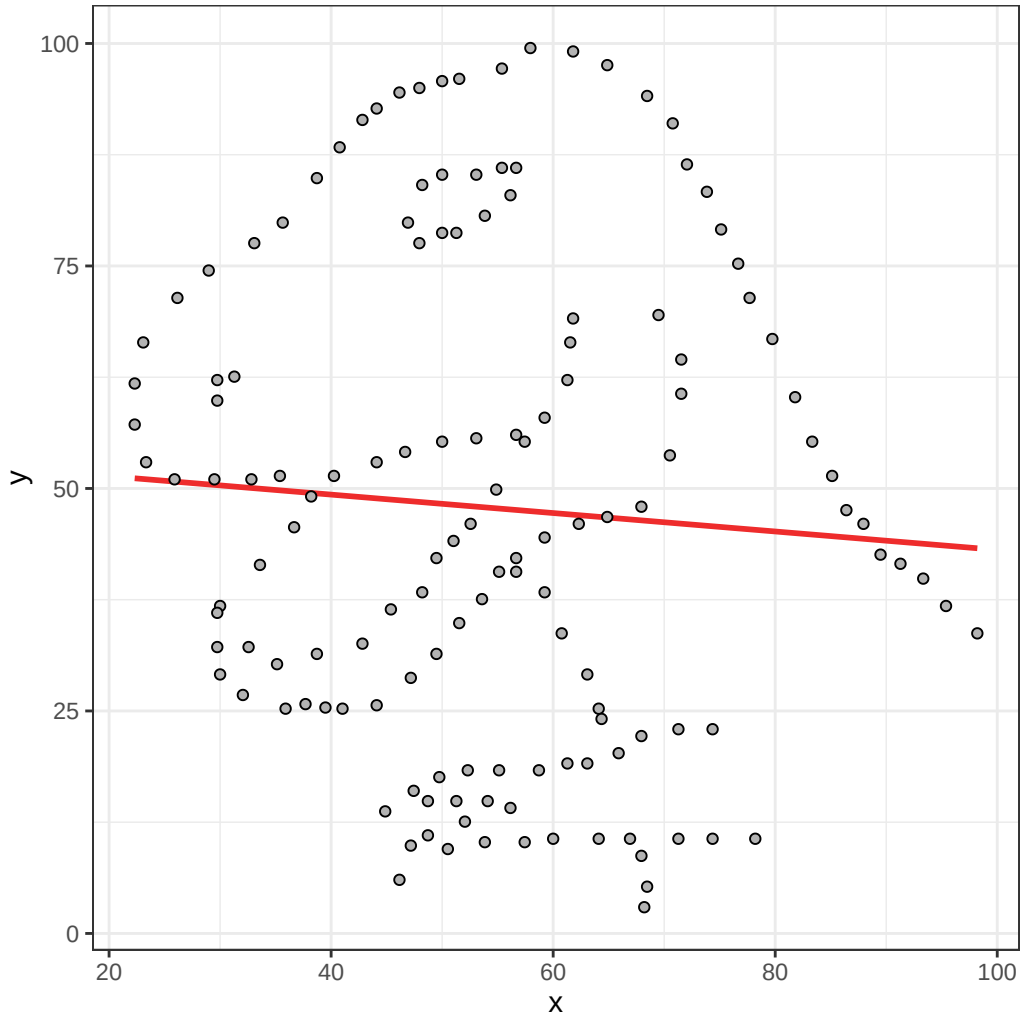
- To complicate things, sometimes assumptions are violated, but the consequences are minor.

Linear Regression Assumptions

$$Y_i = \alpha + \beta \times x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\epsilon^2)$$

1. Linearity
    - The association between x and y is a straight line.
2. Assumptions about the errors
    - Normality: Errors follow a normal distribution.
    - Homoscedasticity: Errors have the same variance for all x.
    - Independence: Errors are independent of each other.

Linearity

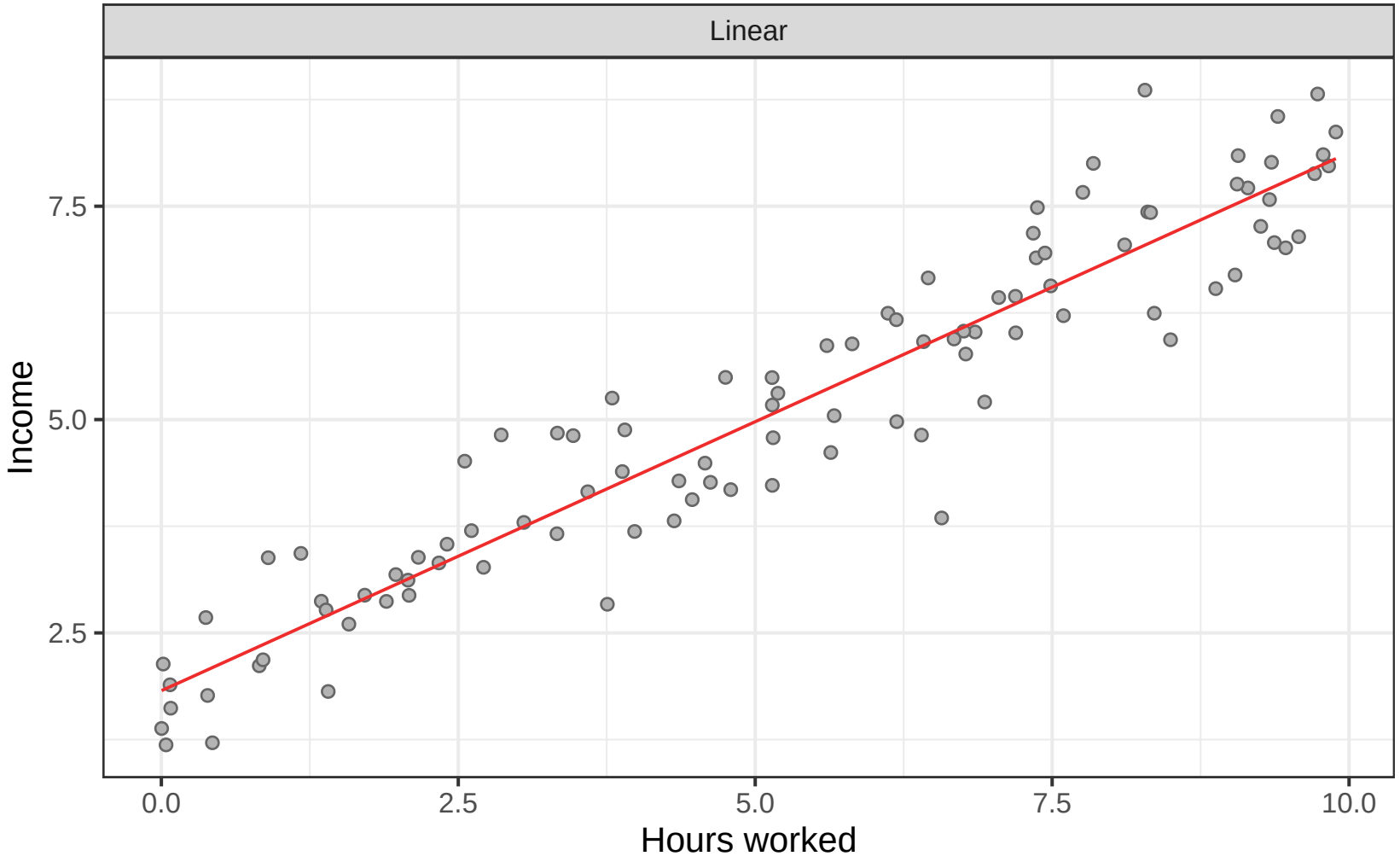


Linearity

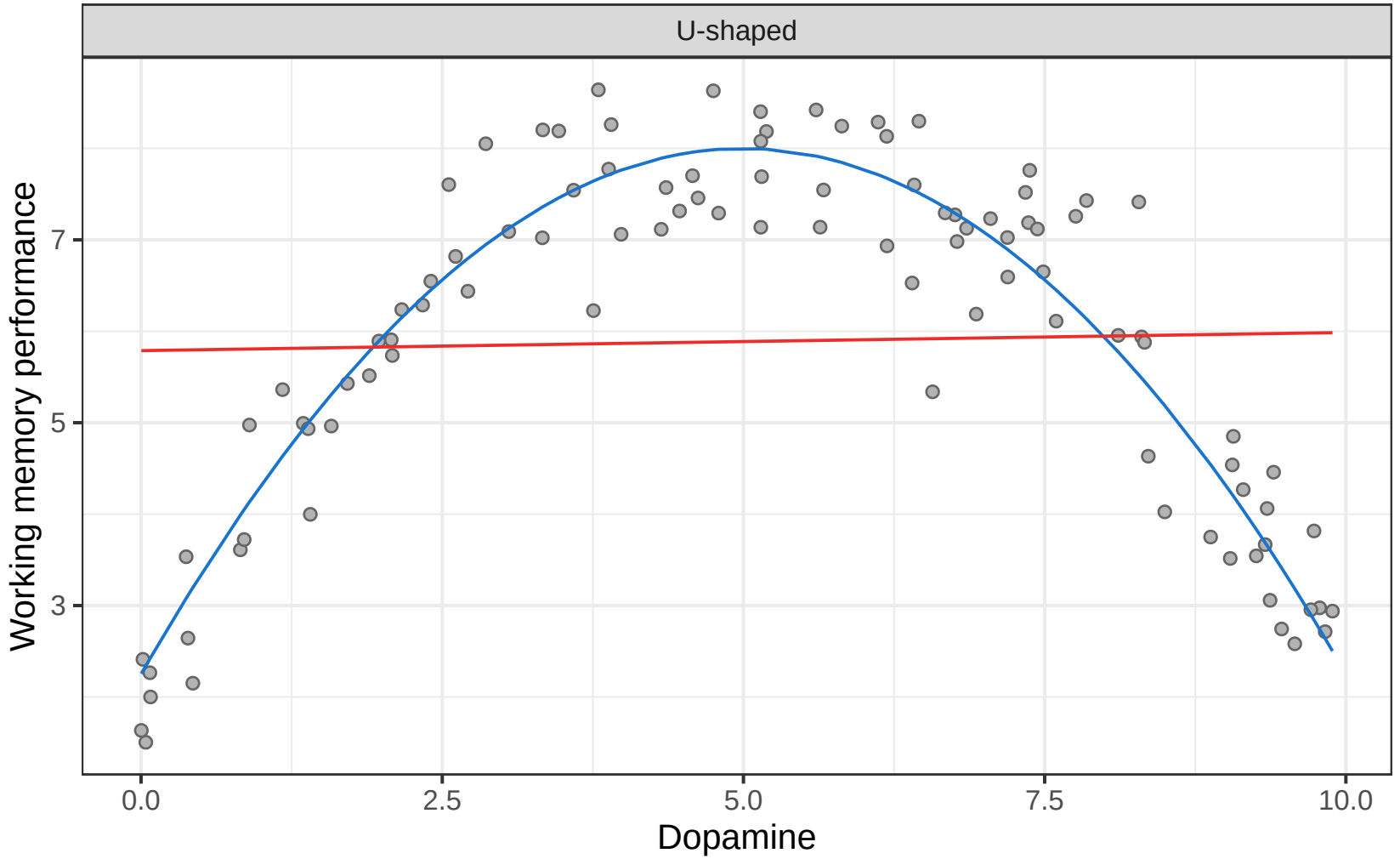
but why might a relationship not be linear?

- Too much of a good thing/ U-shaped
  - Dopamine vs working memory performance
- Diminishing returns/ plateau
  - Study time vs. exam performance
- Threshold effect
  - Swimming lessons vs. distance swum
- Heterogeneous sample

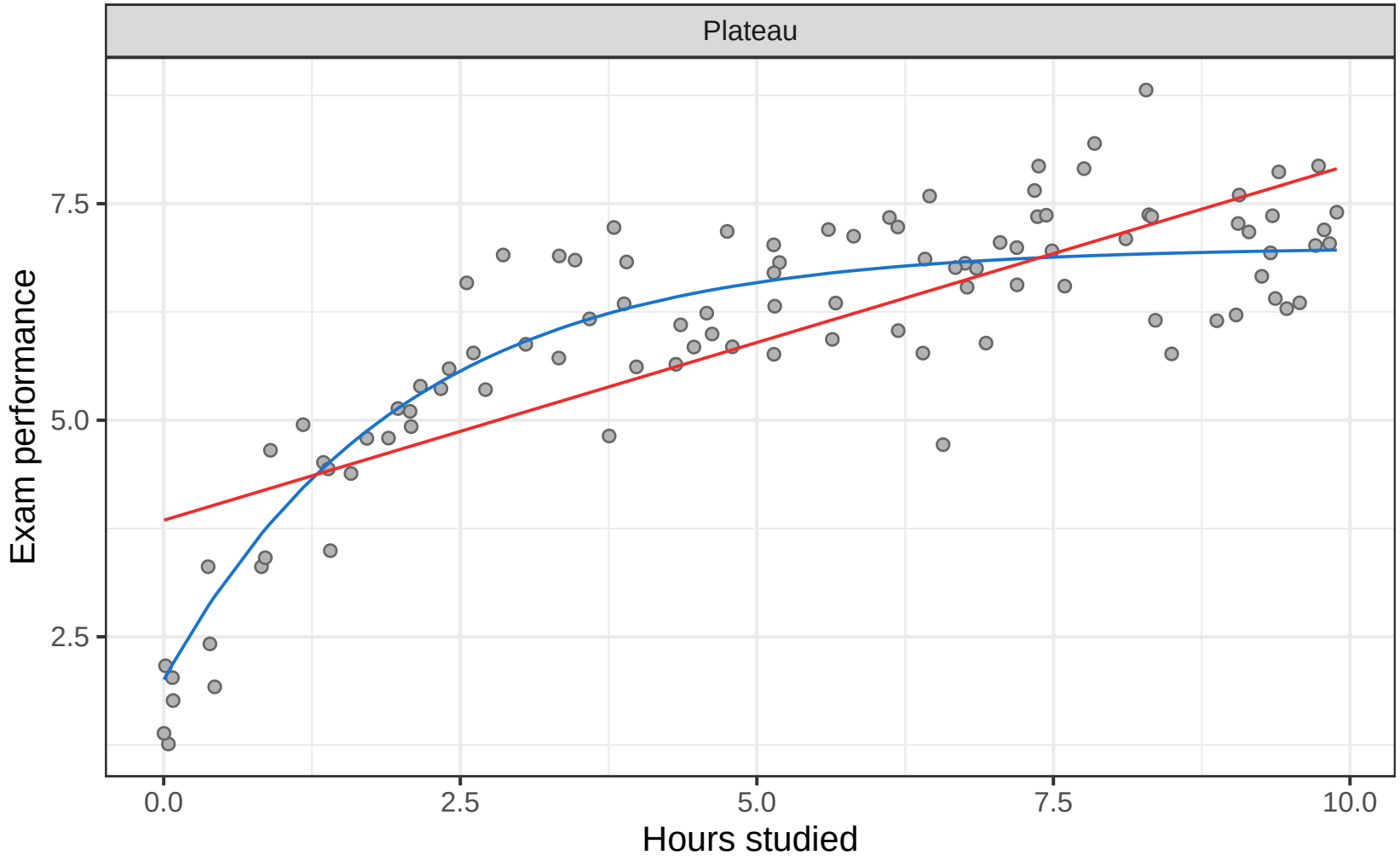
Linearity



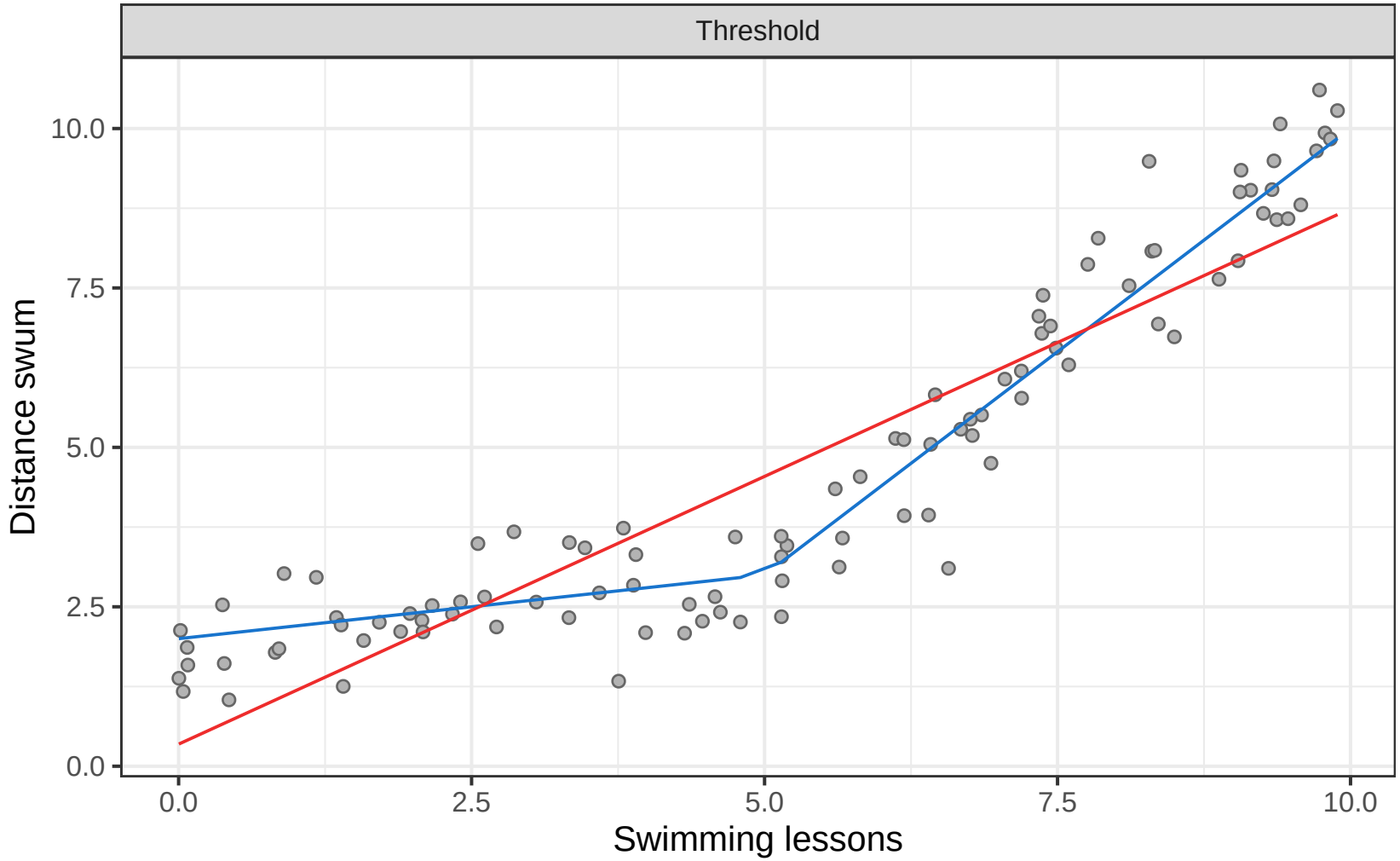
U-shaped


U-shaped
Working memory performance
7
5
3
0.0
2.5
5.0
7.5
10.0
Dopamine

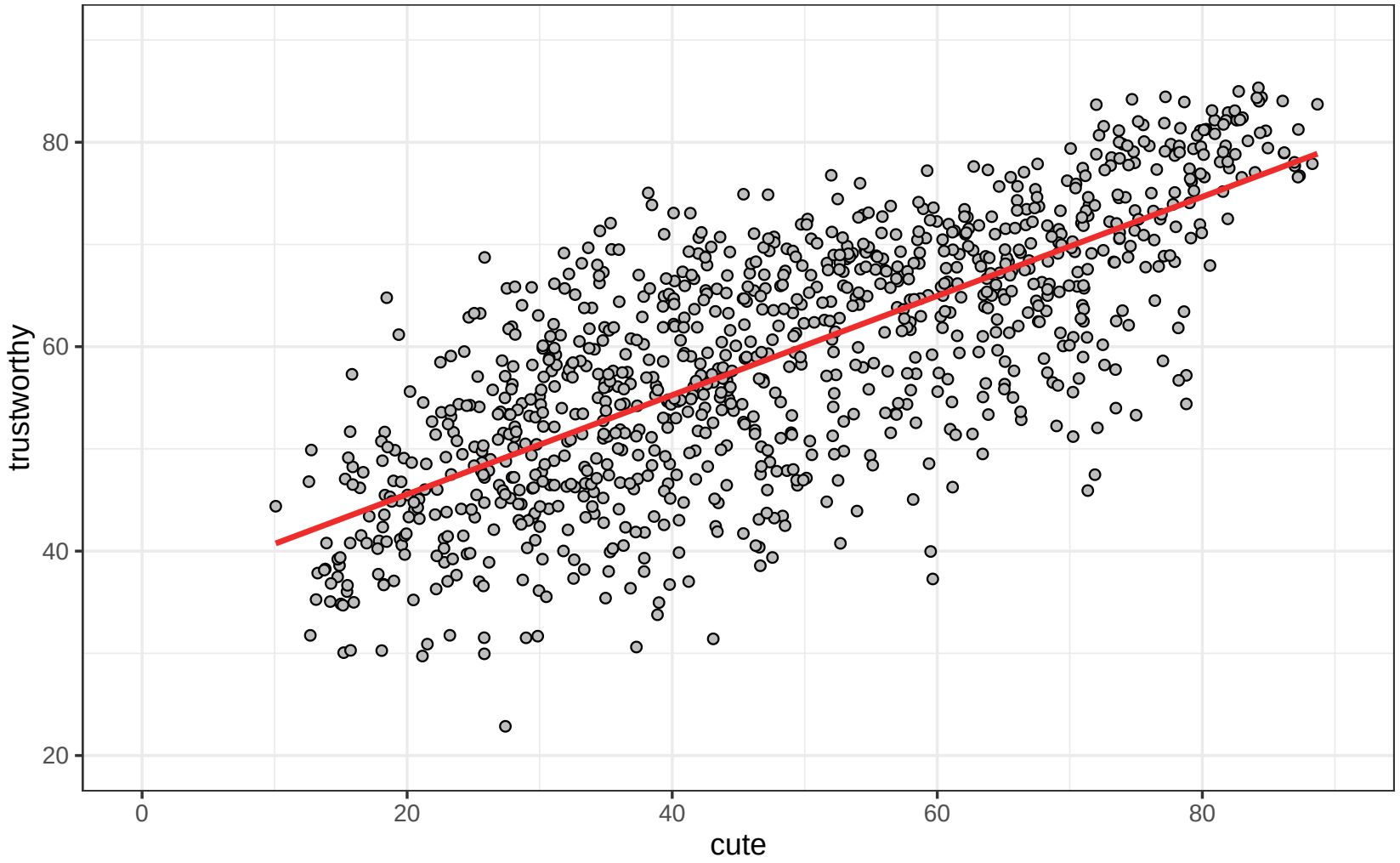
Diminishing returns/ plateau



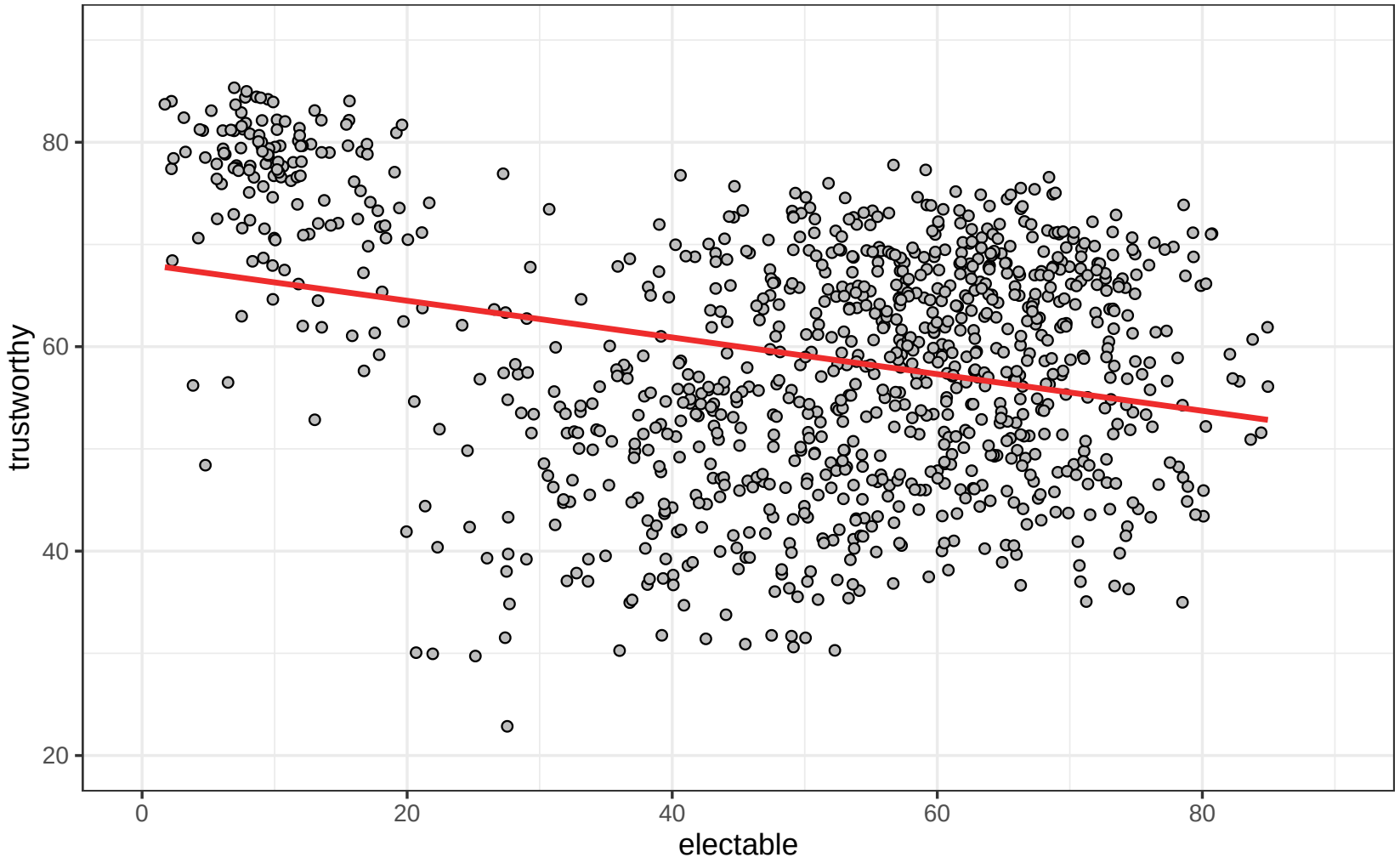
Threshold effect



Faces: Cute vs. Trustworthy



Faces: Electable vs. Trustworthy



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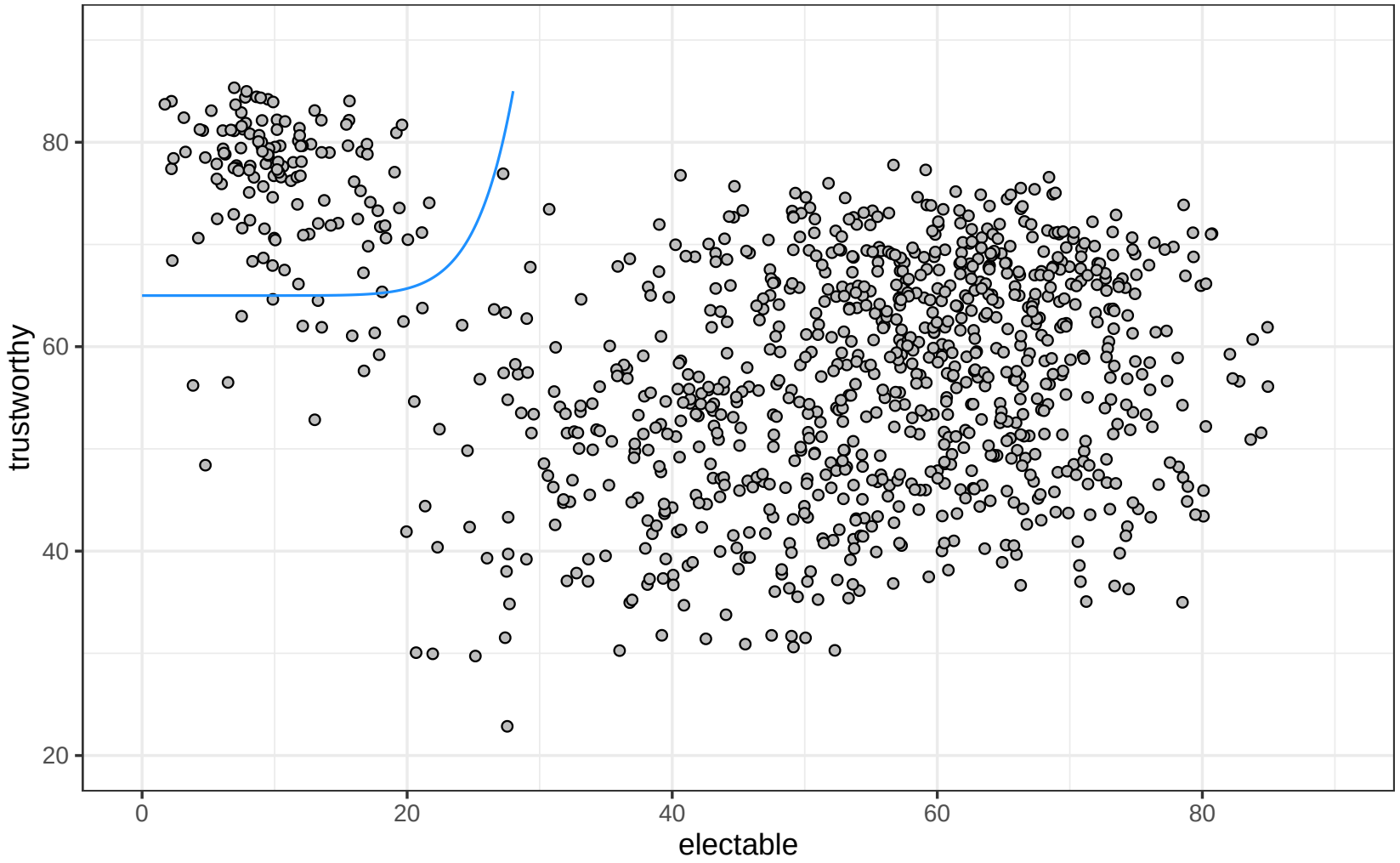

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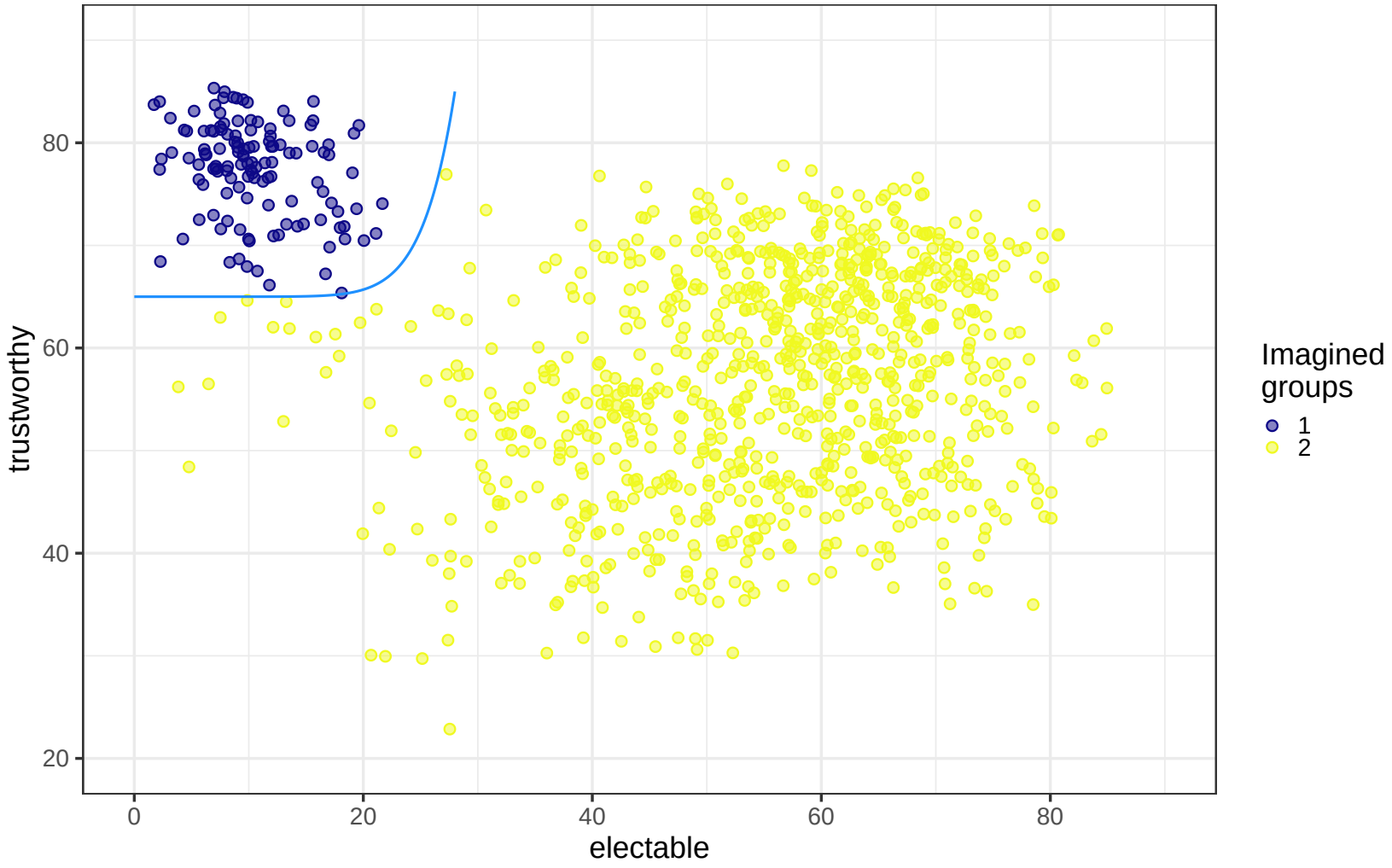
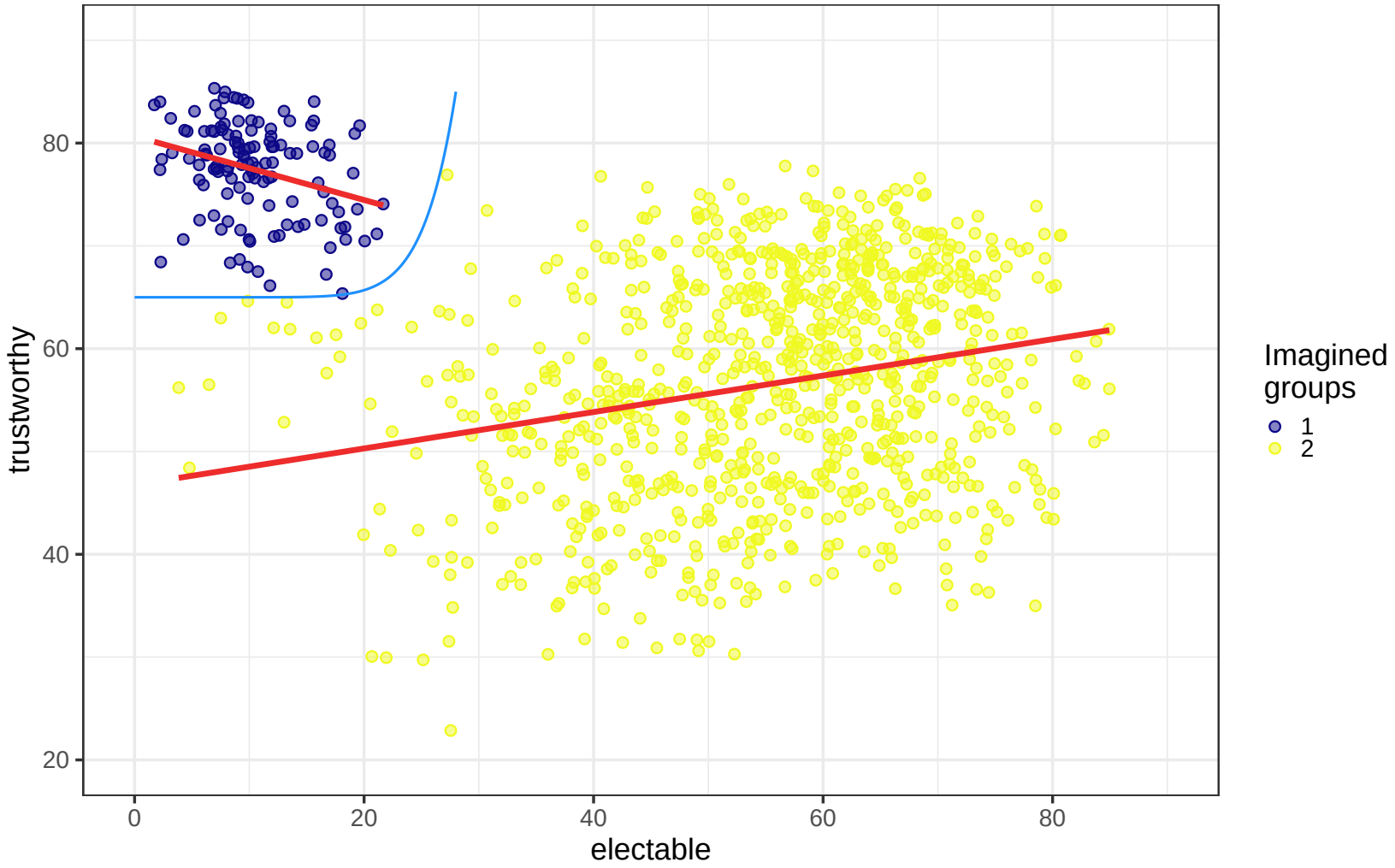


Faces: Electable vs. Trustworthy



Today

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Errors and residuals

$$Y_i = \alpha + \beta \times x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\epsilon^2)$$

Errors and residuals

In the population:

$$Y_i = \alpha + \beta \times x_i + \epsilon_i \qquad \text{so,} \qquad \epsilon_i = Y_i - (\alpha + \beta \times x_i)$$

- The error is the distance from the *population line*.
- We do not know that line, so we cannot observe its errors.

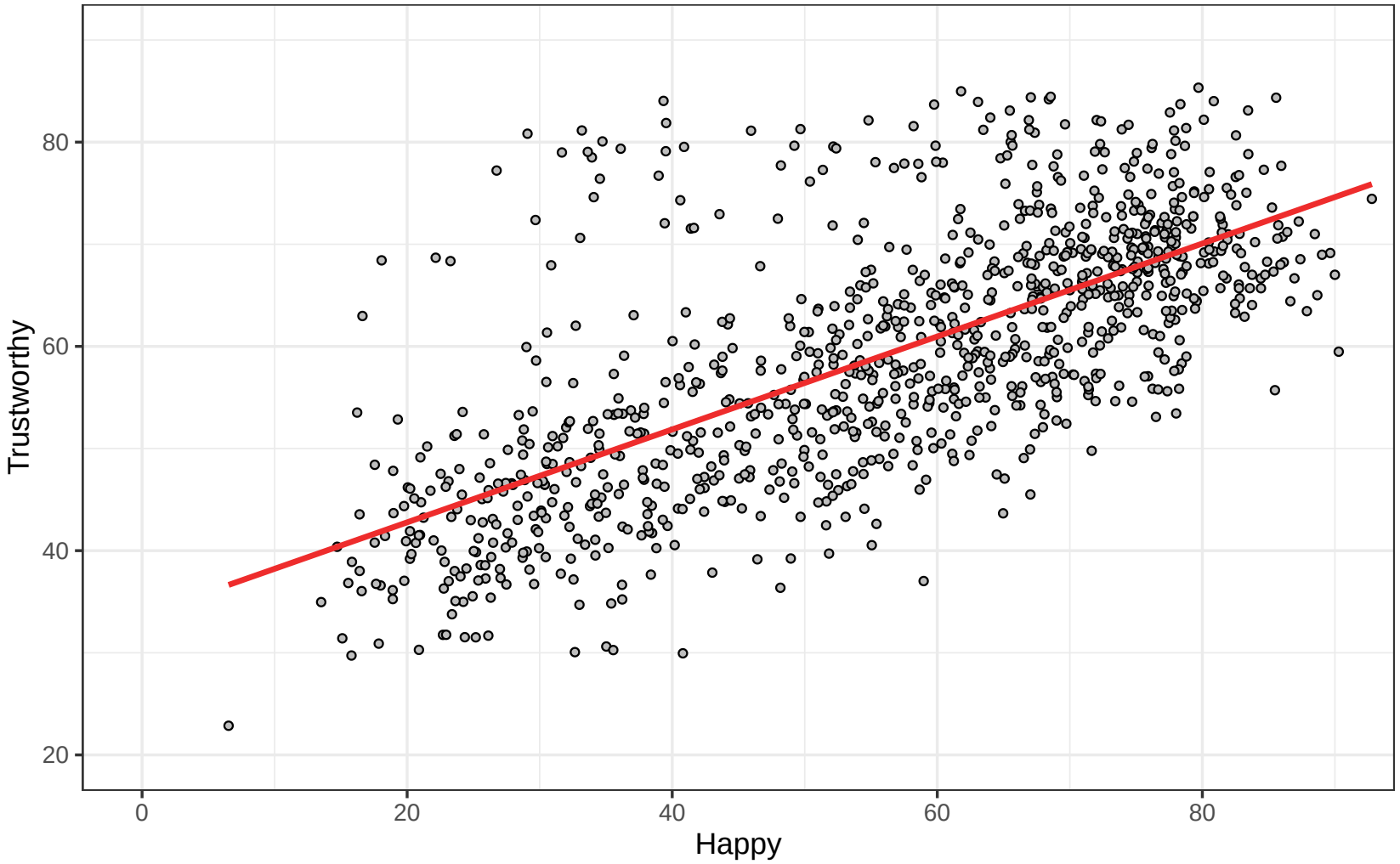
From errors to residuals

In our sample:

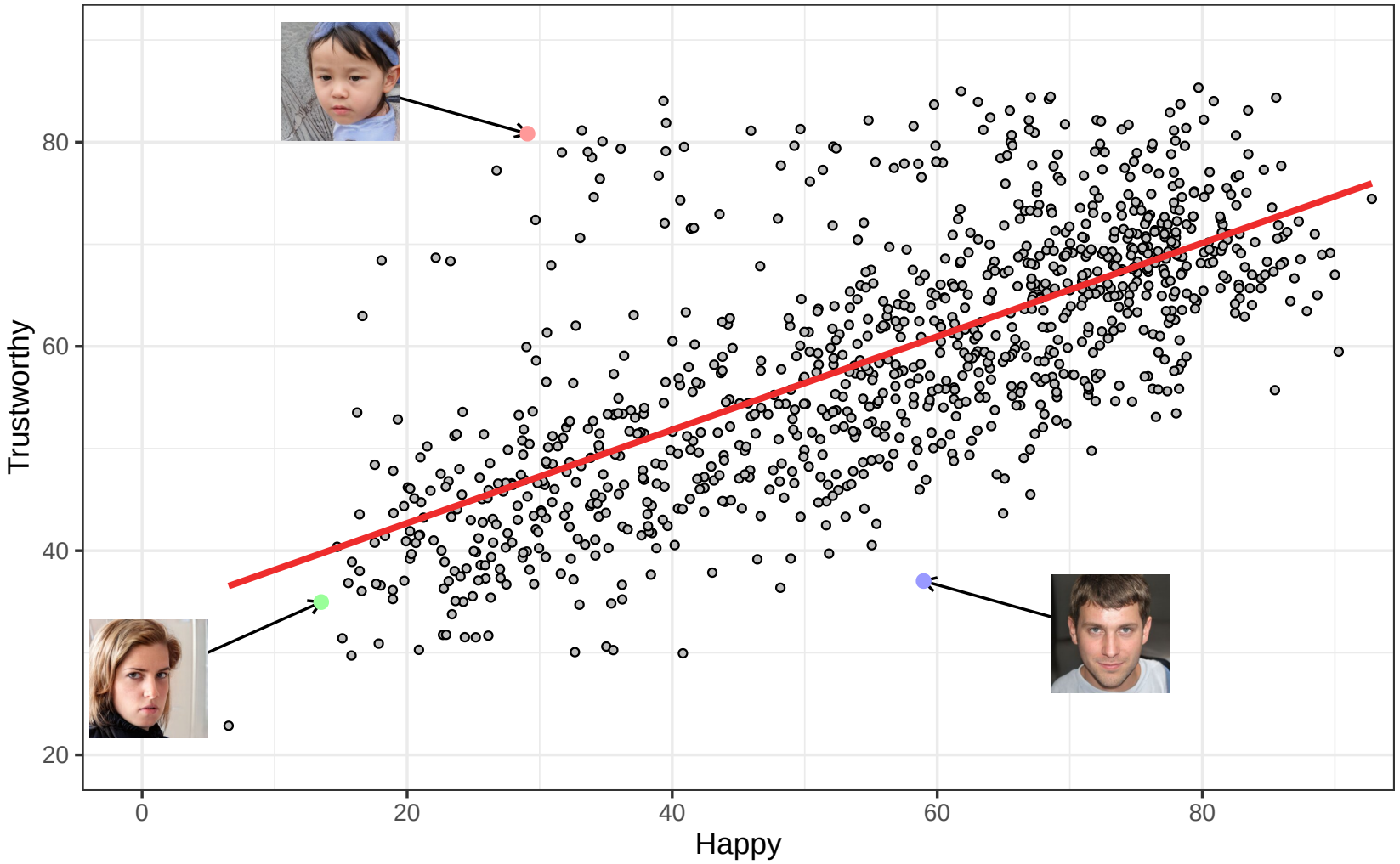
$$\hat{y}_i = a + b \times x_i \qquad \text{so,} \qquad e_i = y_i - \hat{y}_i = y_i - (a + b \times x_i)$$

- The residual is the distance from the *fitted line*.
- Residuals help us assess the assumptions about errors.

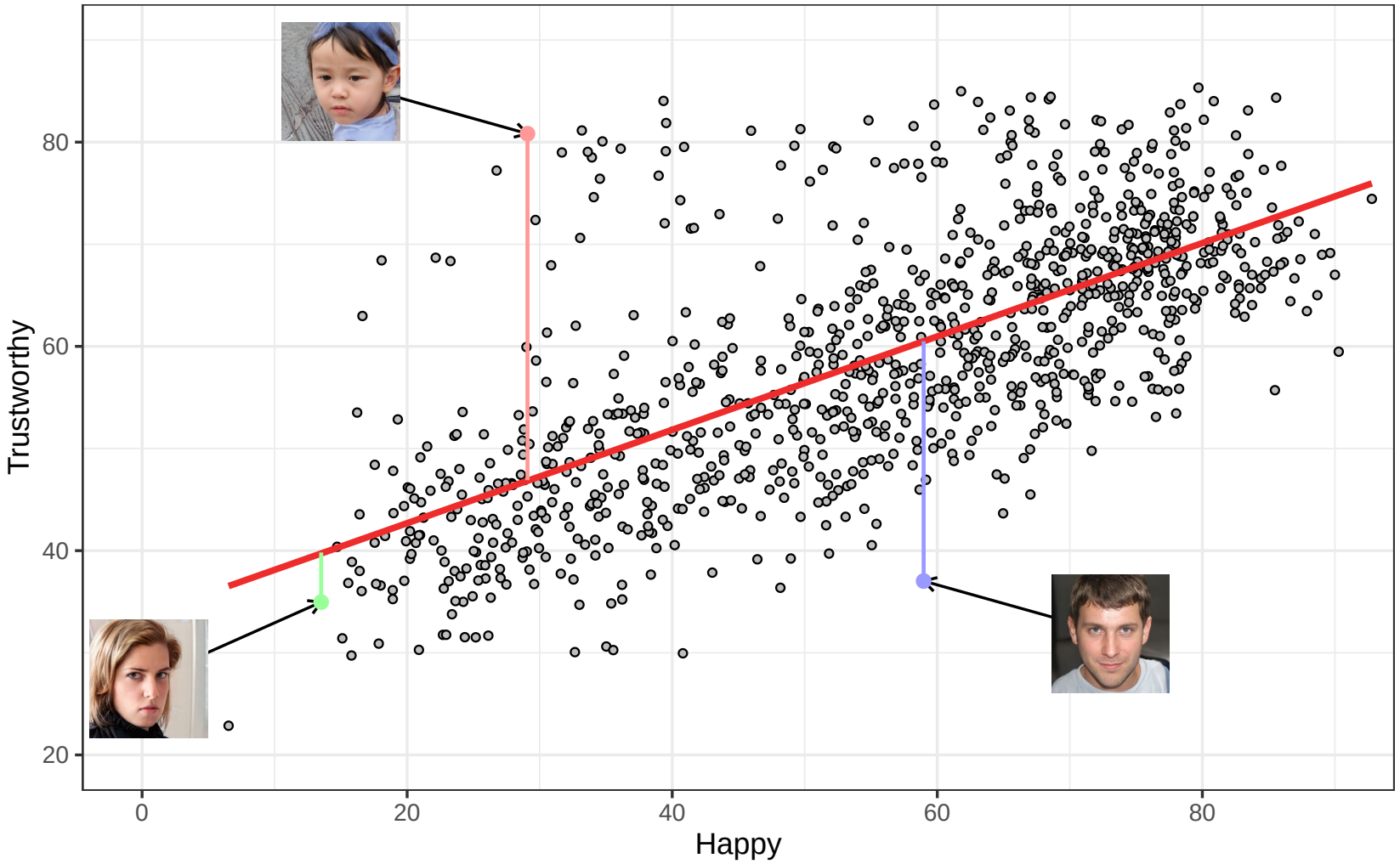
From the regression line to residuals



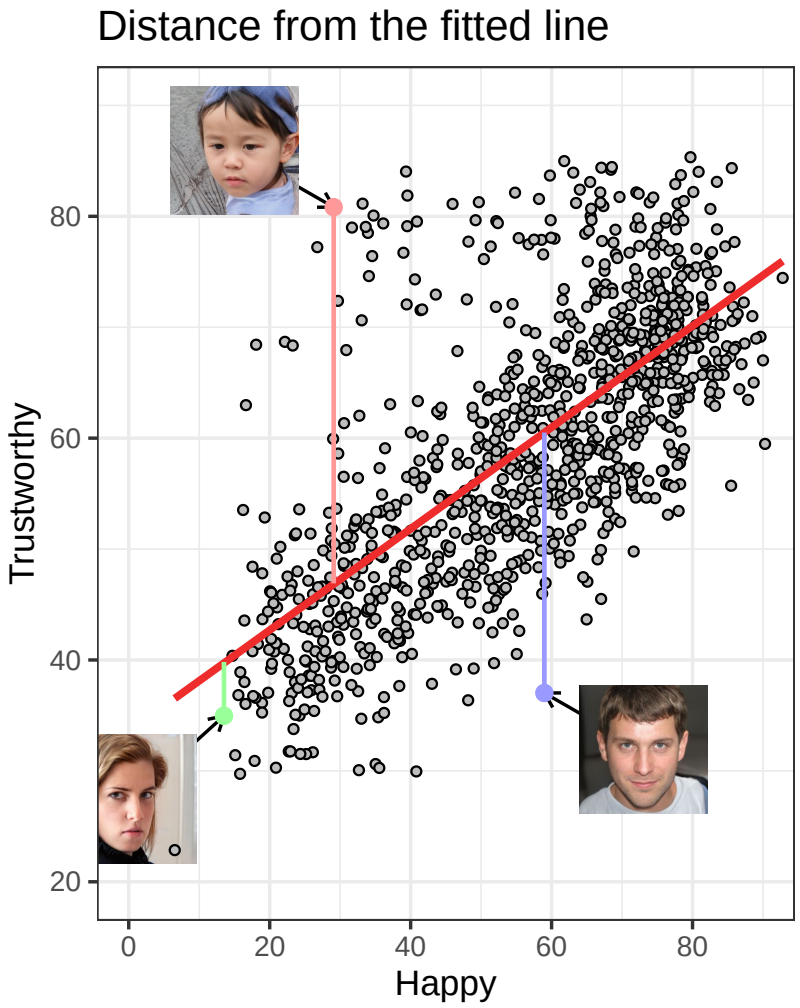
From the regression line to residuals

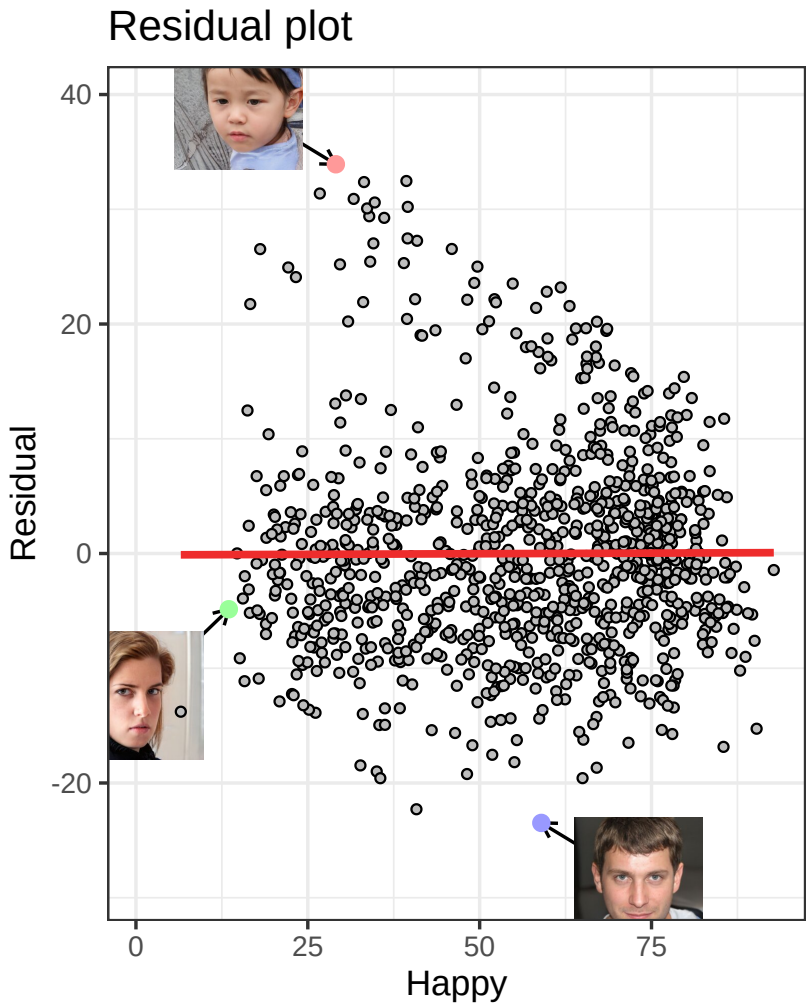


From the regression line to residuals



From the regression line to residuals





From the regression line to residuals

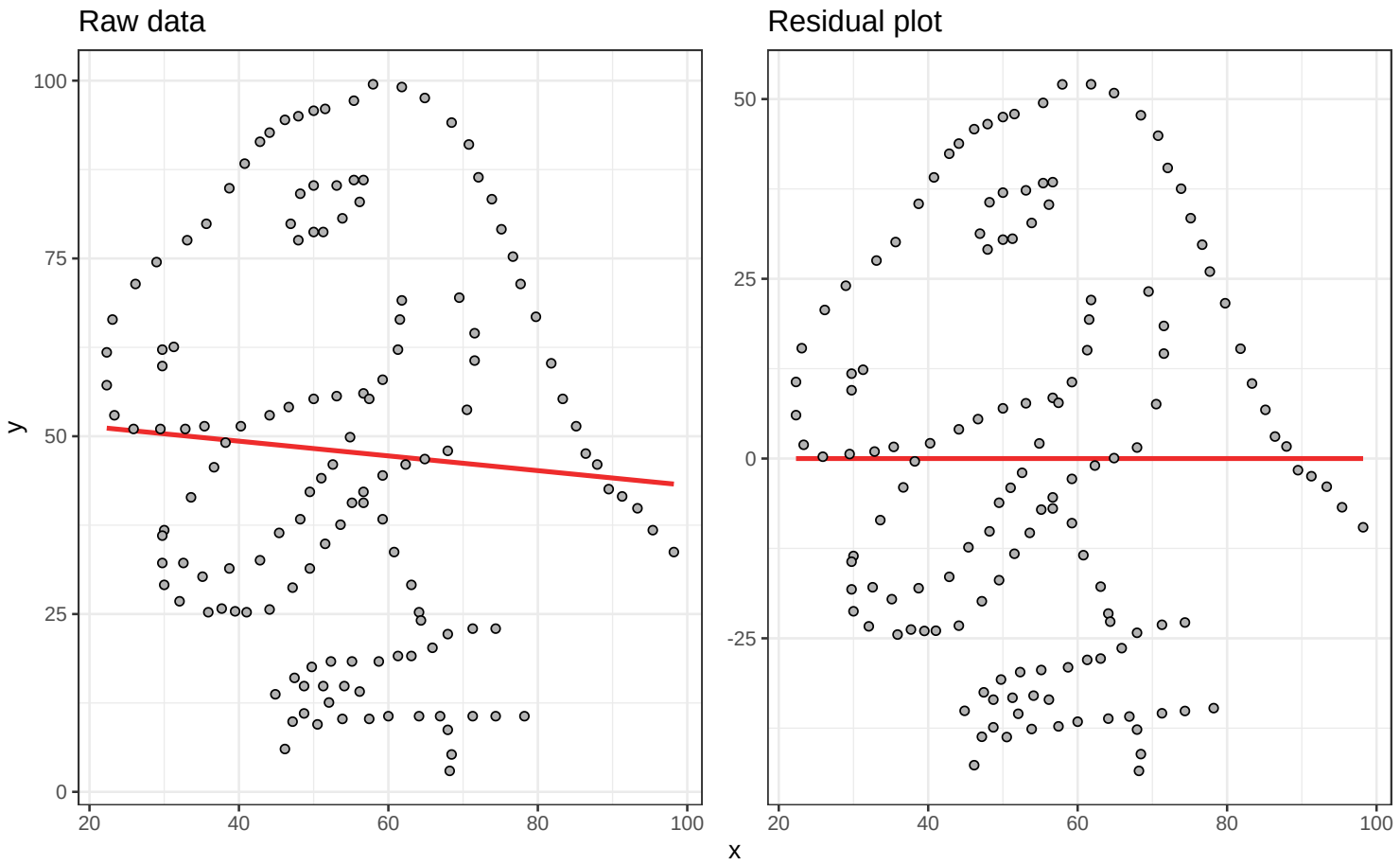
- Keep each face’s happy rating on the horizontal axis.
- Replace its trustworthy rating with its residual.
- The colored points were the same observation in each panels.

What are we looking for?

We can do (almost) all assumption checks using the residuals

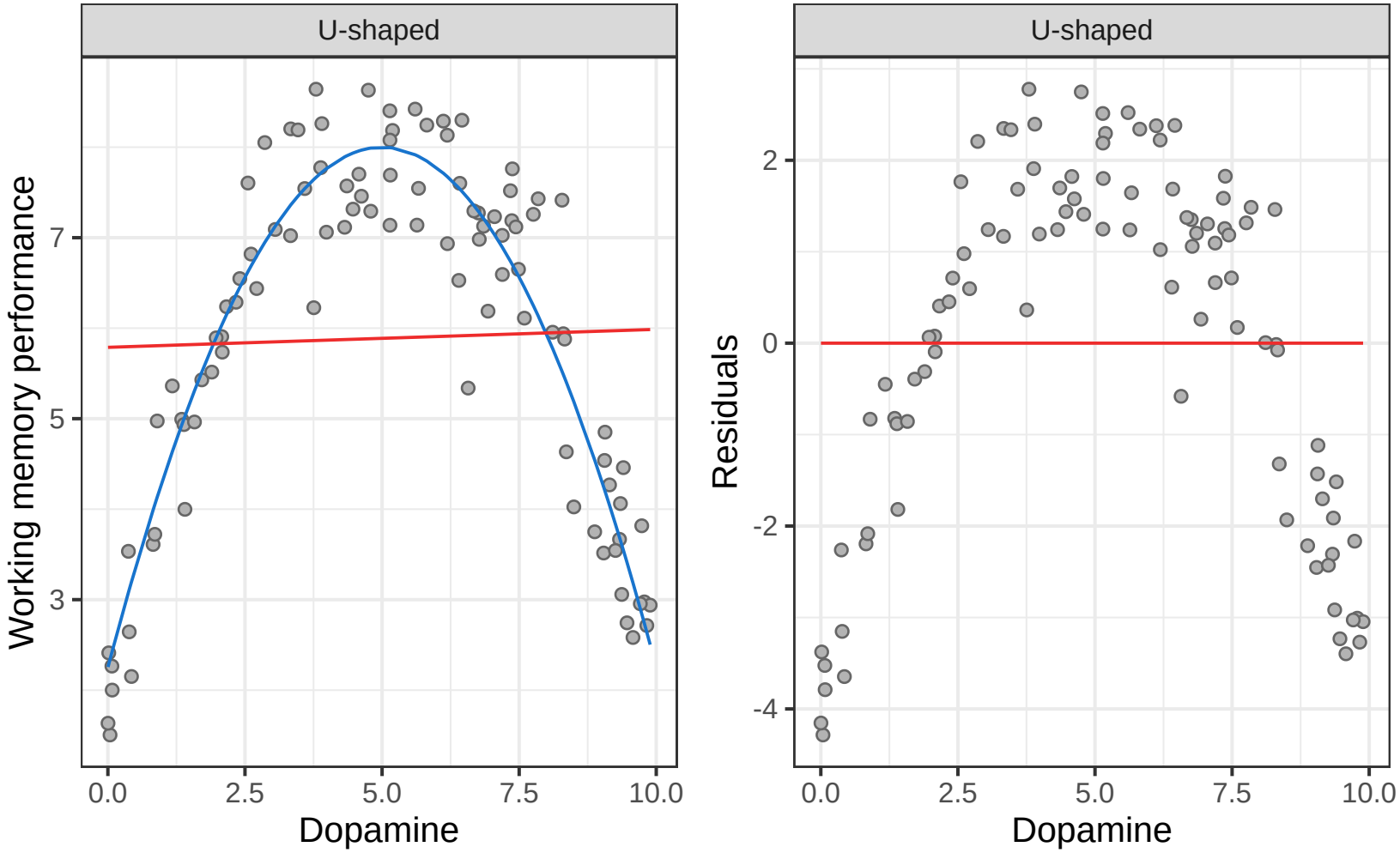
- If there is a systematic pattern, this suggests that the regression line misses part of the relationship.
- That is a check for linearity!

Checking linearity from the residuals

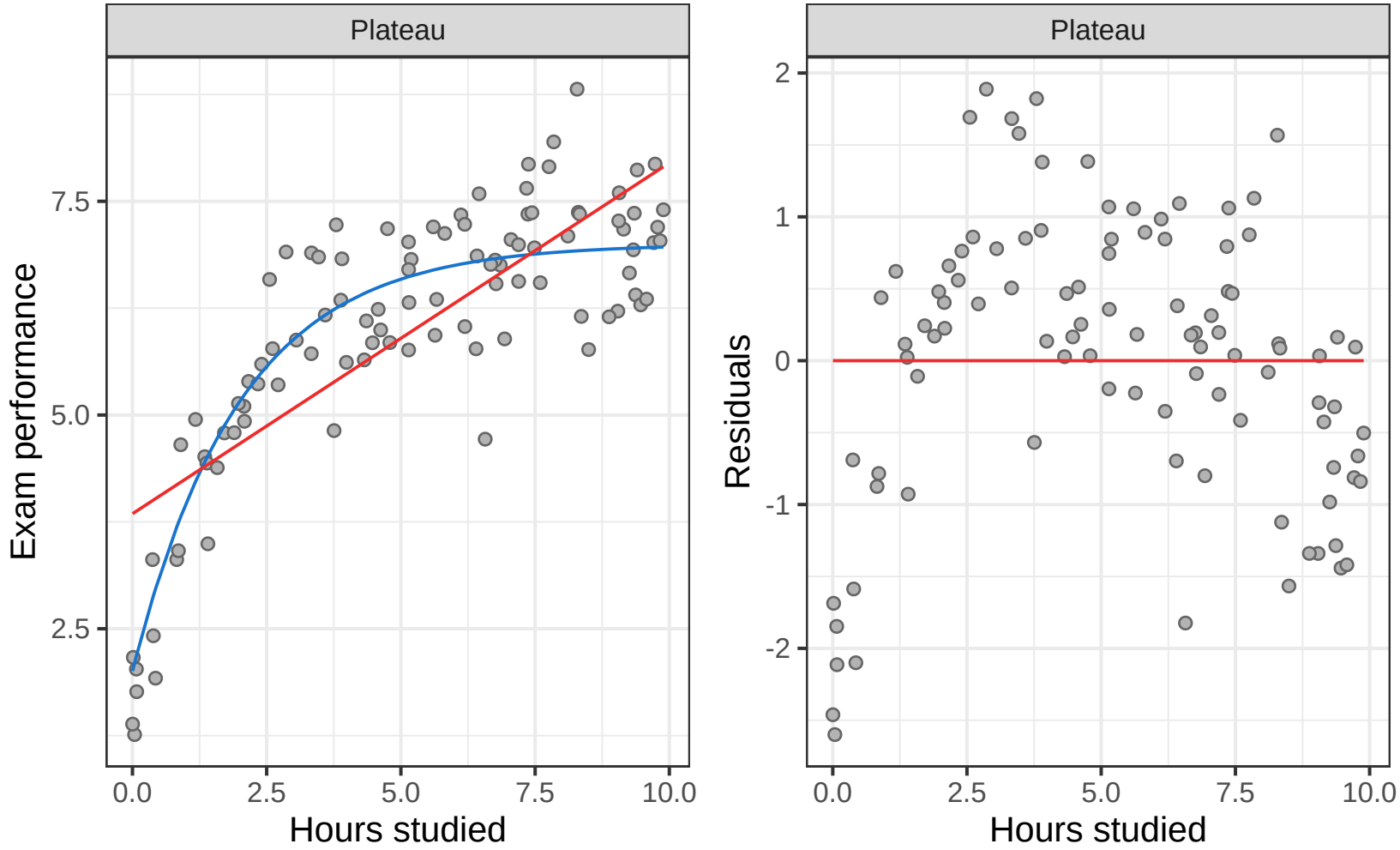


The dino is the nonlinear signal.

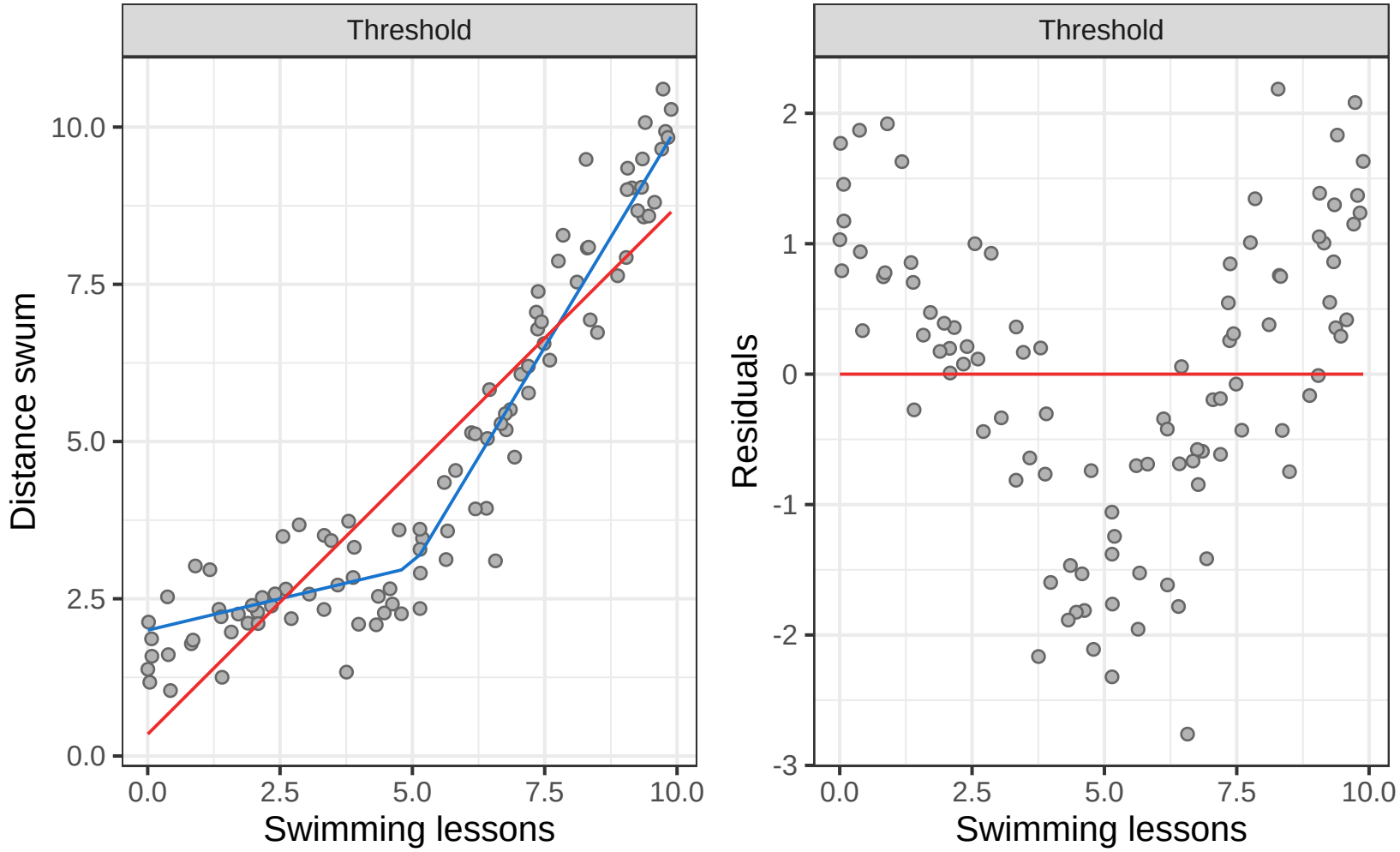
Checking linearity from the residuals



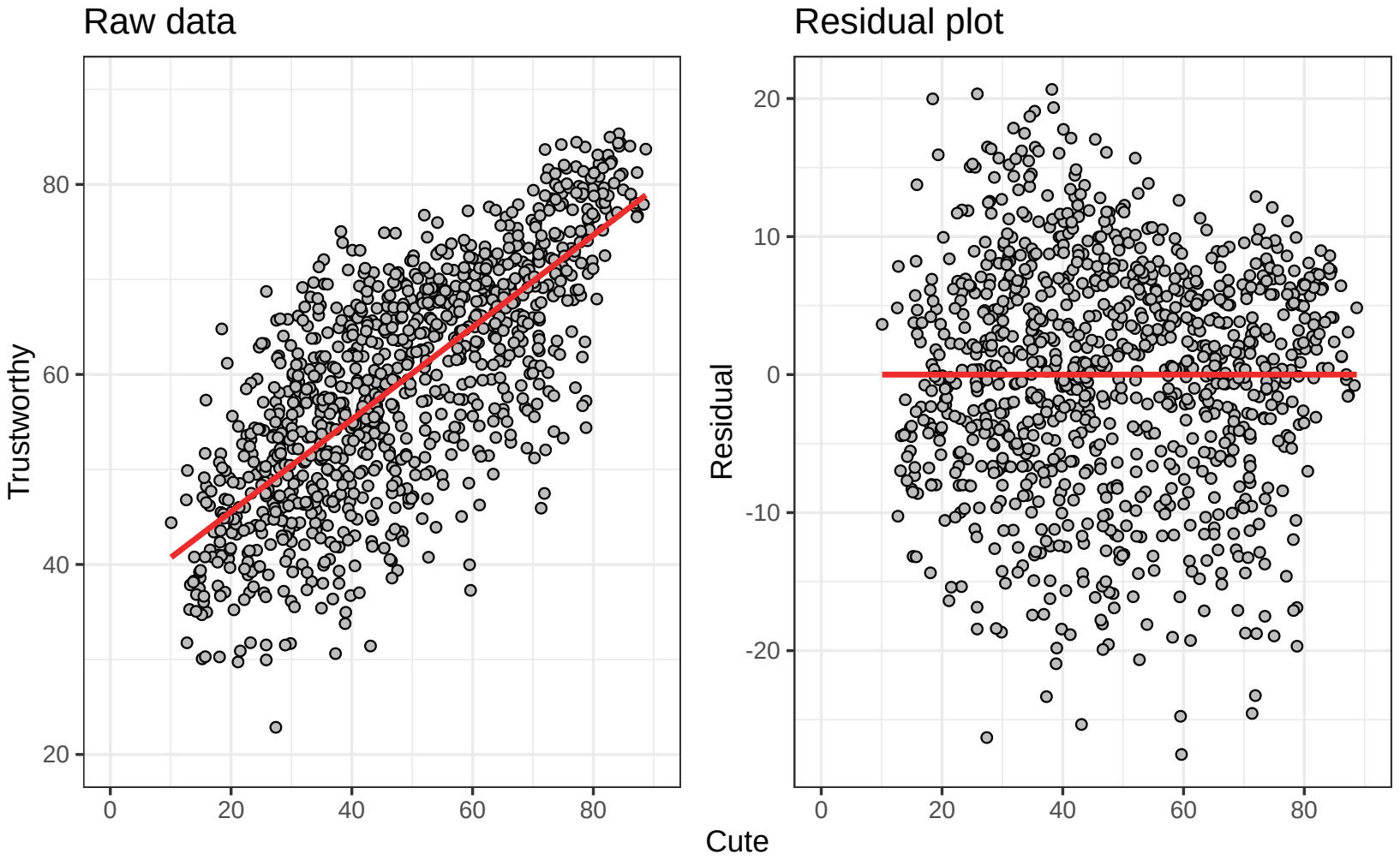
Checking linearity from the residuals



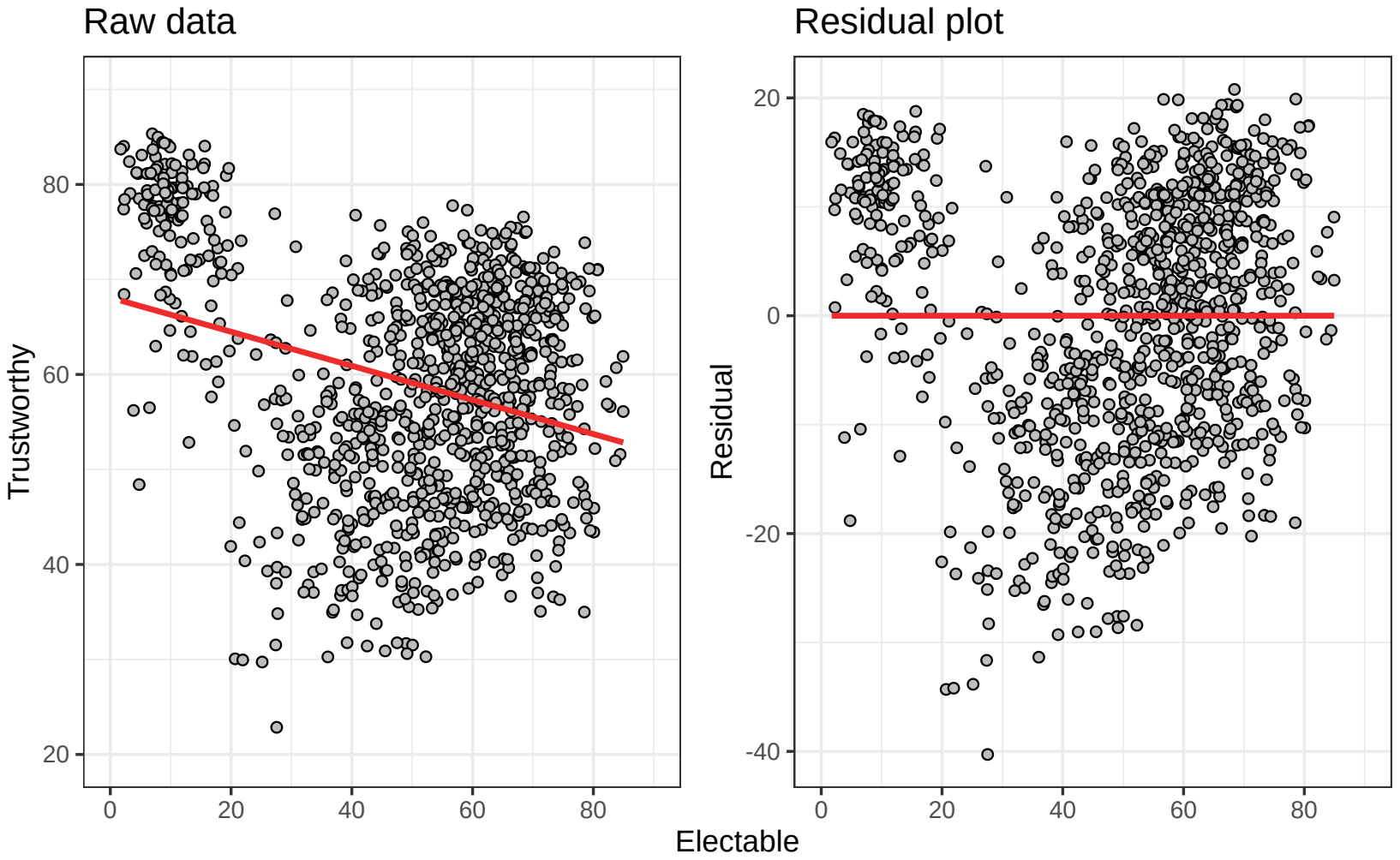
Checking linearity from the residuals



Faces: Cute vs. Trustworthy



Faces: Electable vs. Trustworthy



Residuals and Linearity

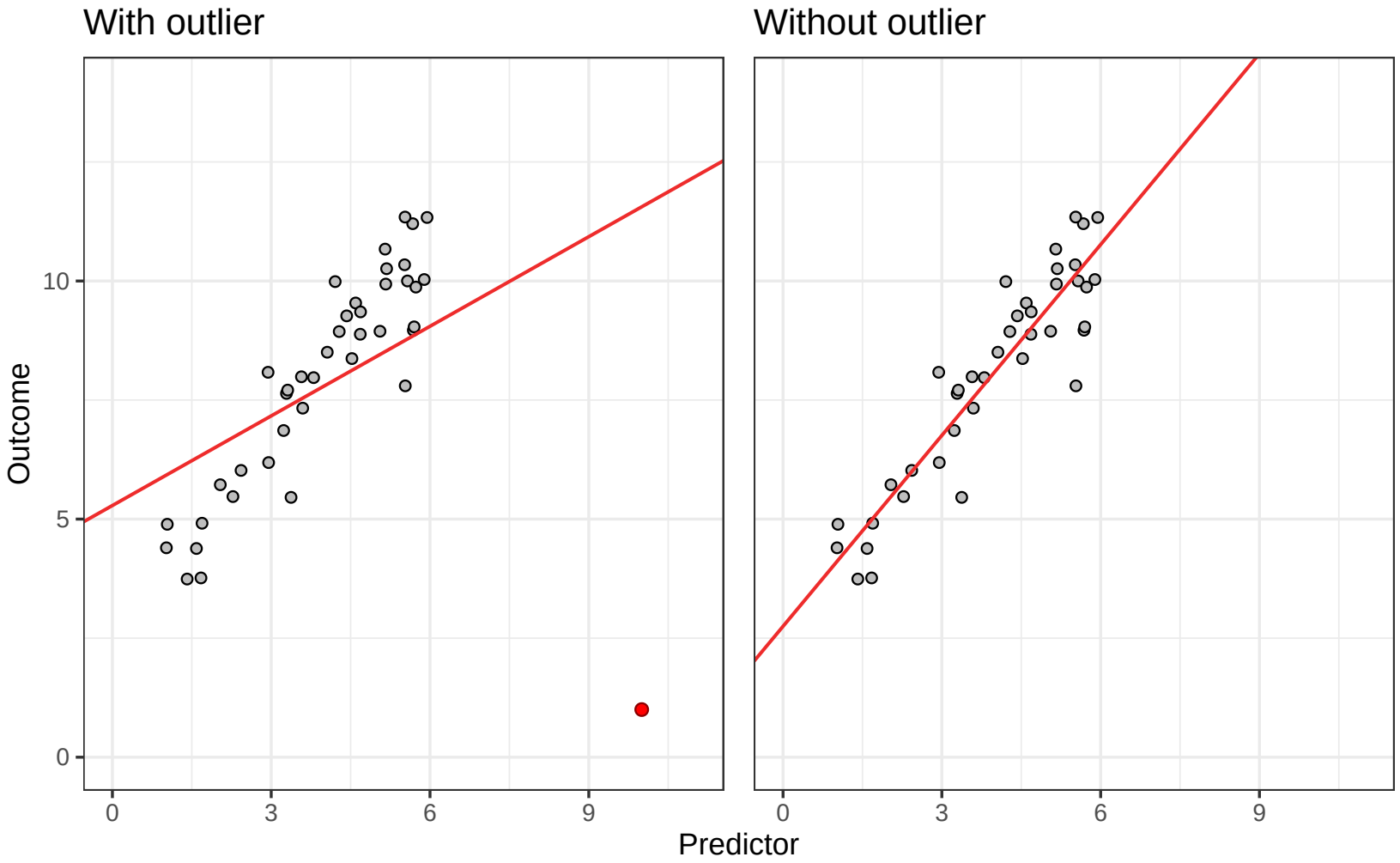
- We can examine the residuals for any systematic pattern.
  - If we find one, this can suggest that the regression line misses part of the relationship.
    - the part that is nonlinear.

Sometimes there is not a pattern, but just one or a few outlying observations.

These are called *outliers*.

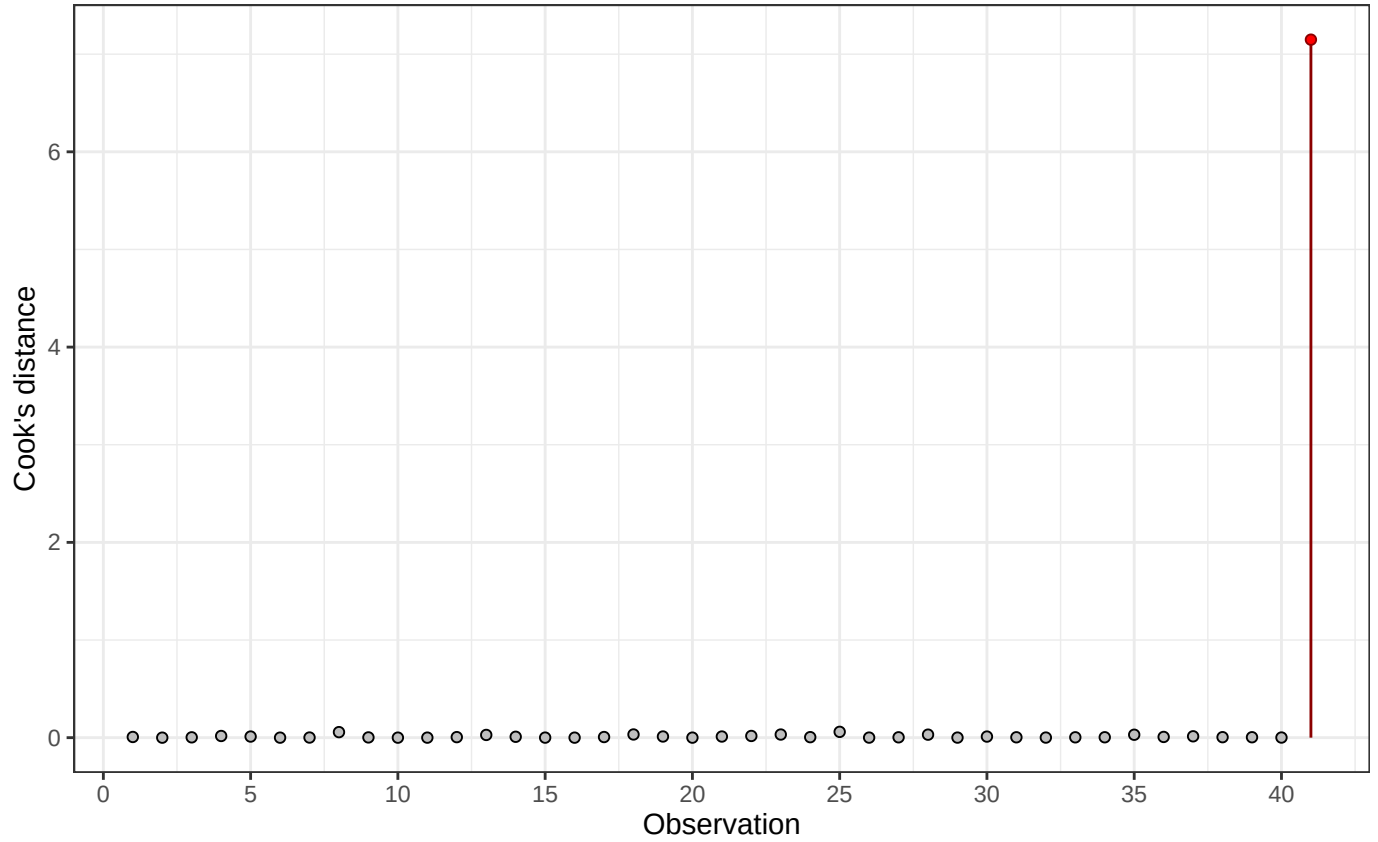
- Outliers can have a large impact on the regression line.
- But not every outlier strongly influences the regression.
- We investigate what happens when we leave out an observation.

Outliers



Detecting outliers: Cook's distance

Cook's distance measures how much our fitted regression changes when we leave out one observation.



Detecting outliers: Cook’s distance

- Larger Cook’s distance means a more influential observation.
- Larger Cook’s distances deserve closer inspection.
- Do not automatically remove them!

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Normality

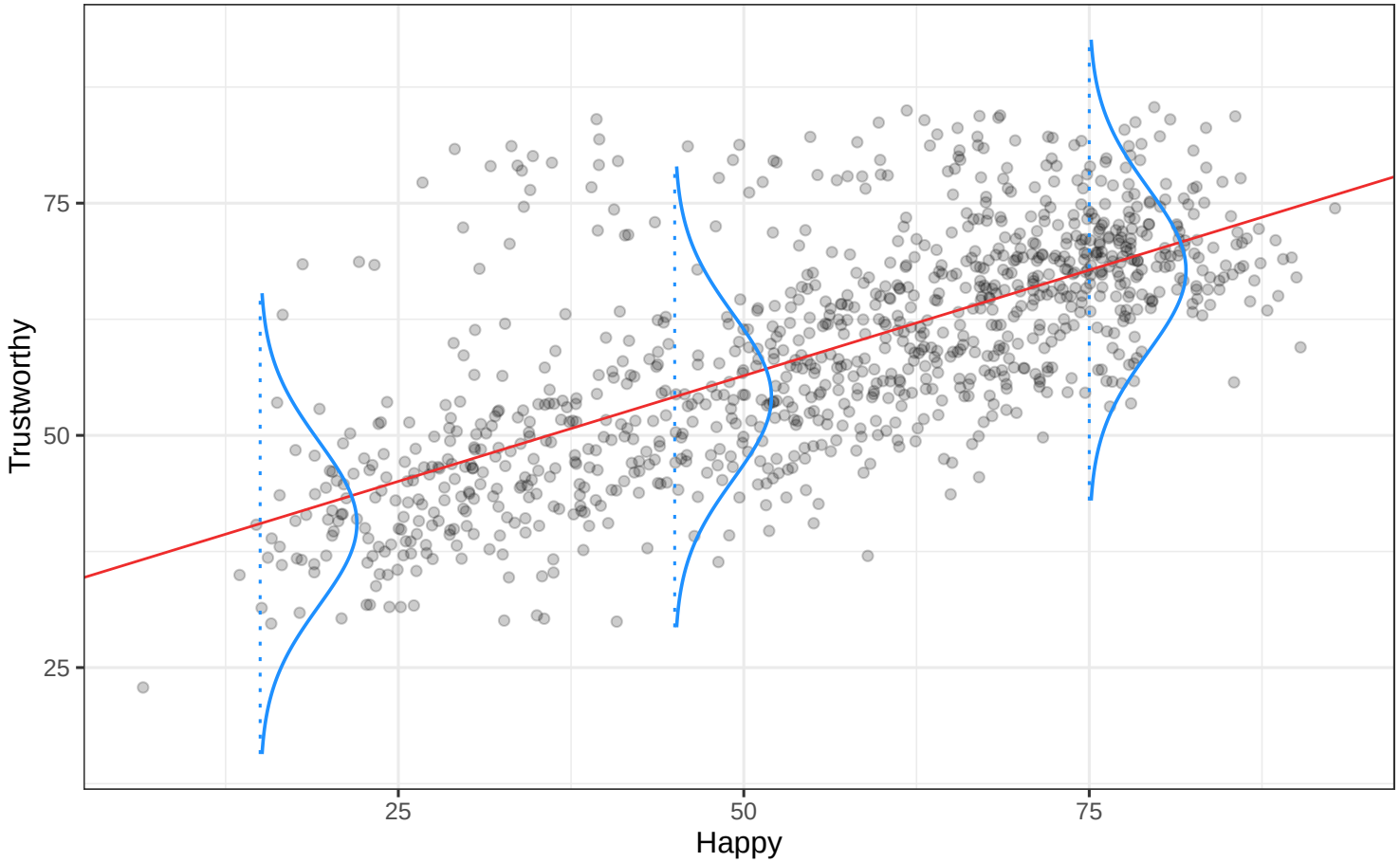
$$Y_i = \alpha + \beta \times x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2_\epsilon)$$

Normality: Errors follow a normal distribution.

- The assumption concerns **errors around the population line**.
- The outcome variable y does not need to be normally distributed.
- Only the prediction error is!

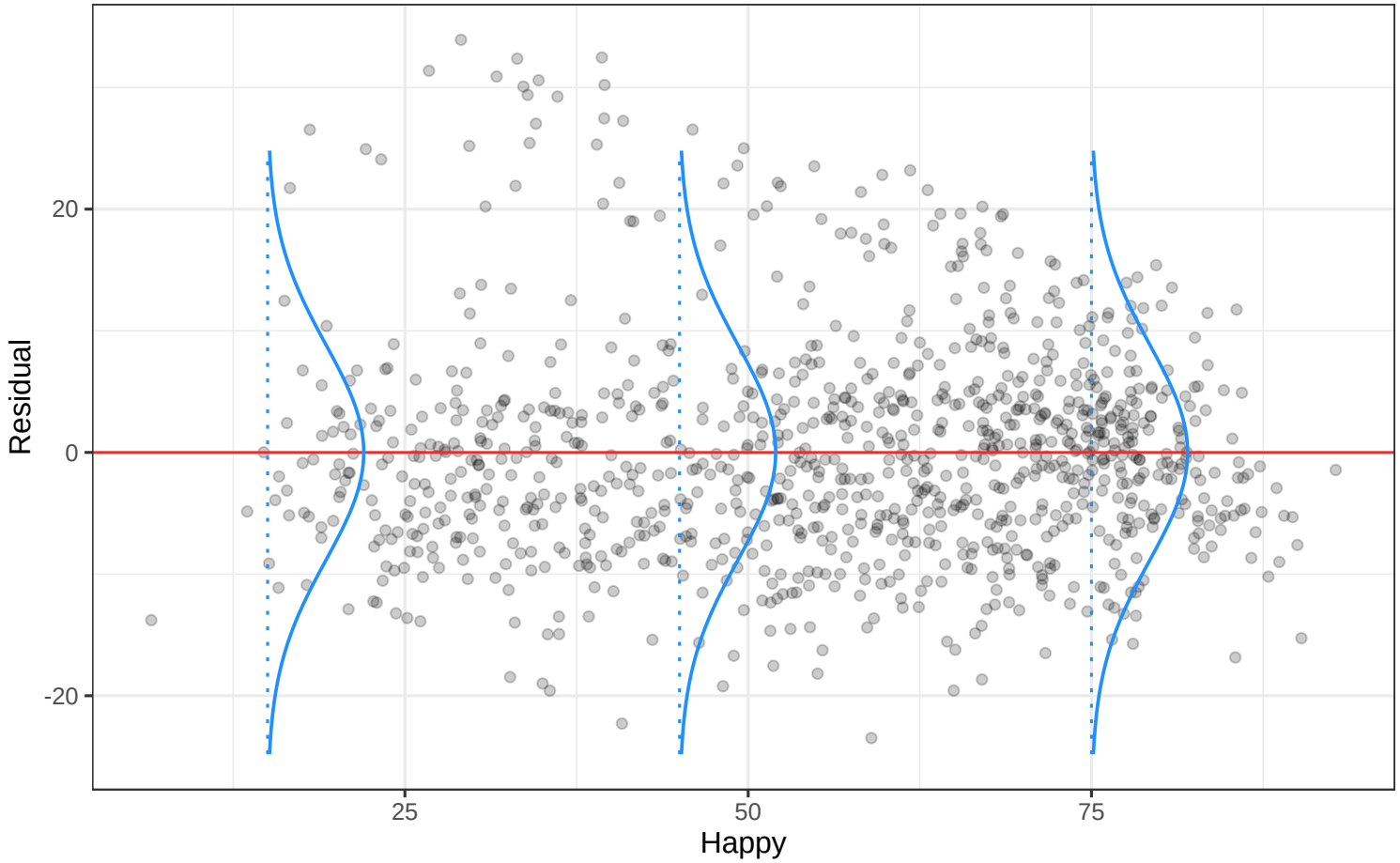
We'll look at some examples and develop a new tool to examine deviations from normality: QQ-plots

Normality



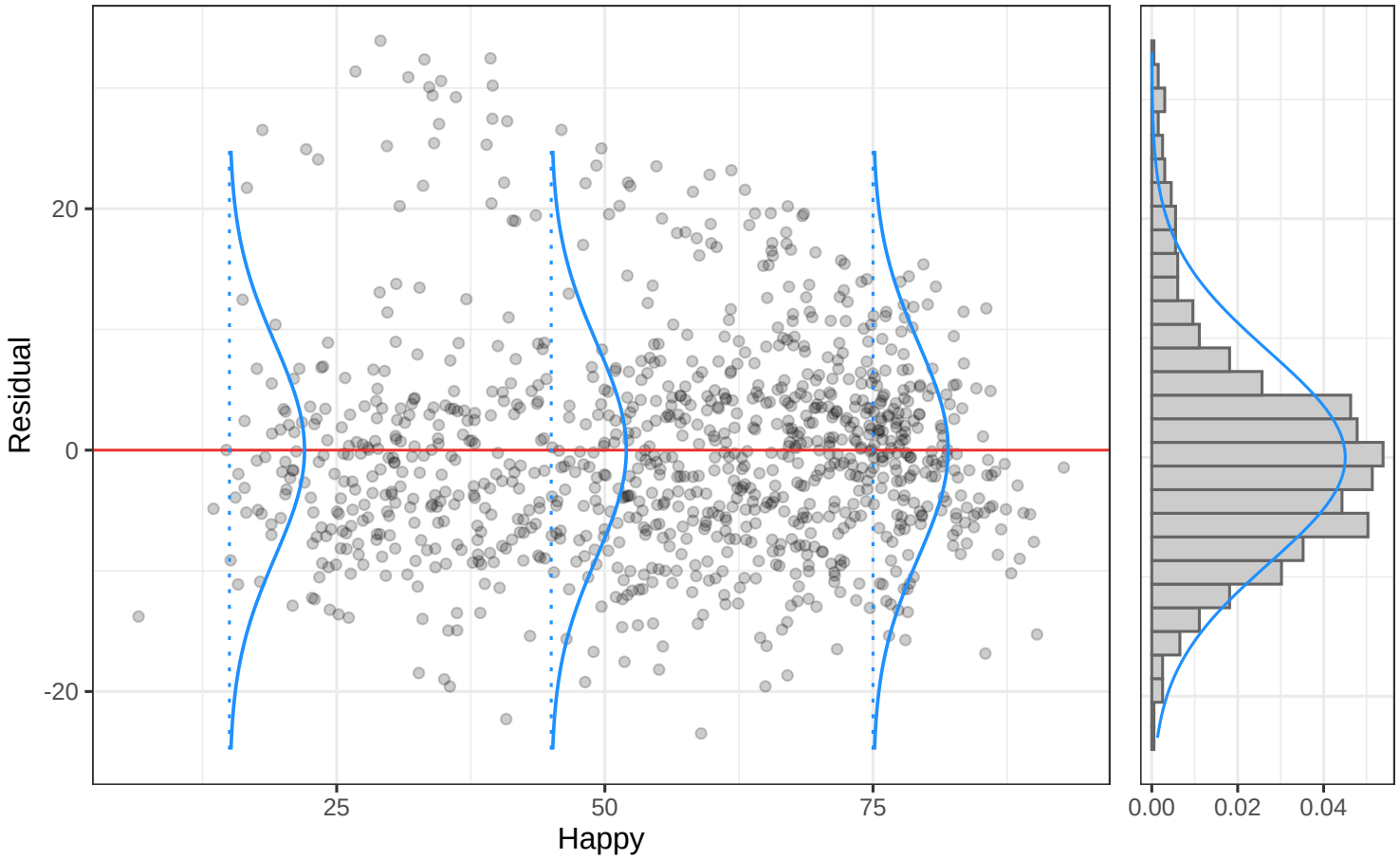
At each value of the predictor x_i, mistakes are normally distributed.

Normality



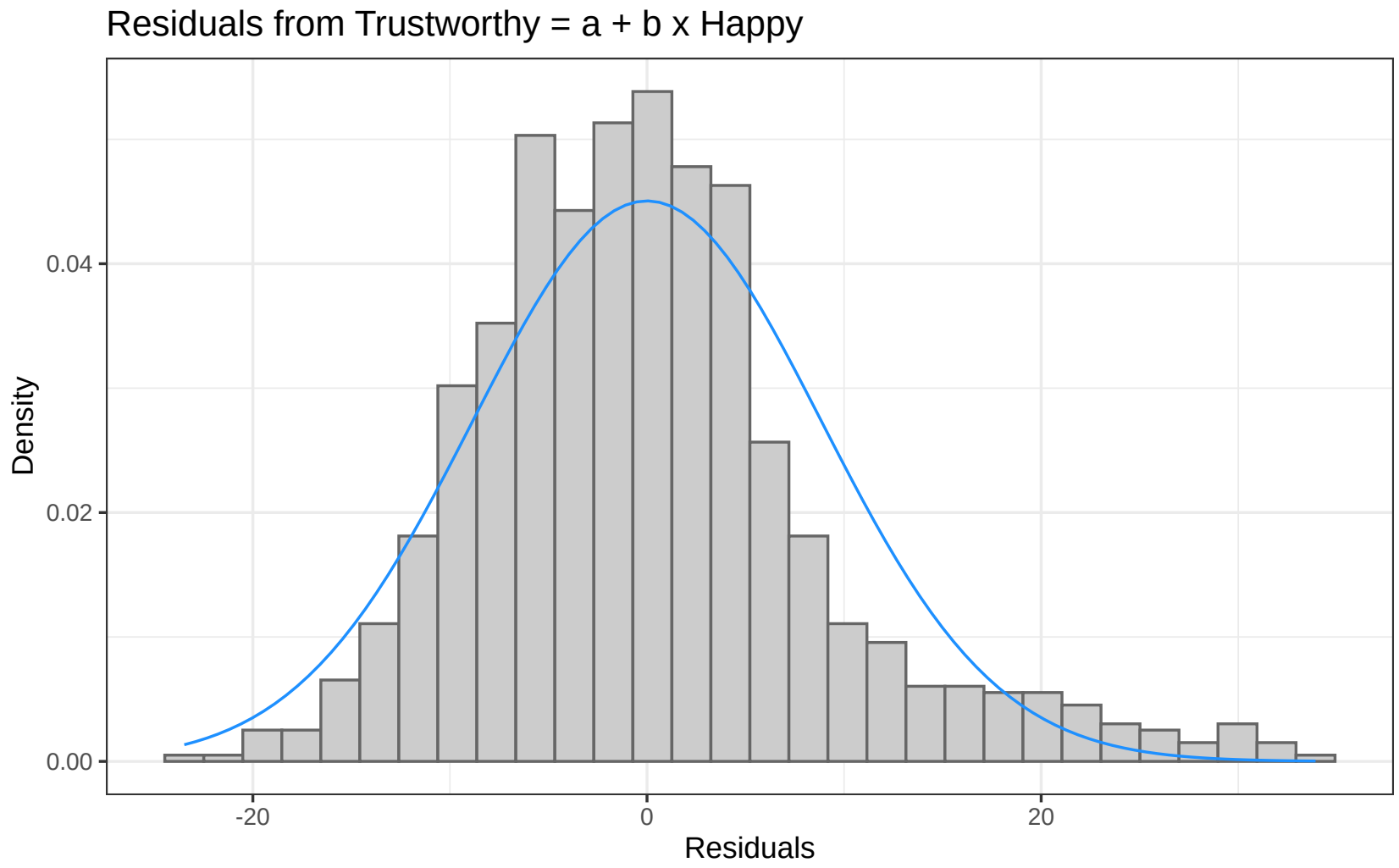
By examining the residuals, the trend disappears.

Normality

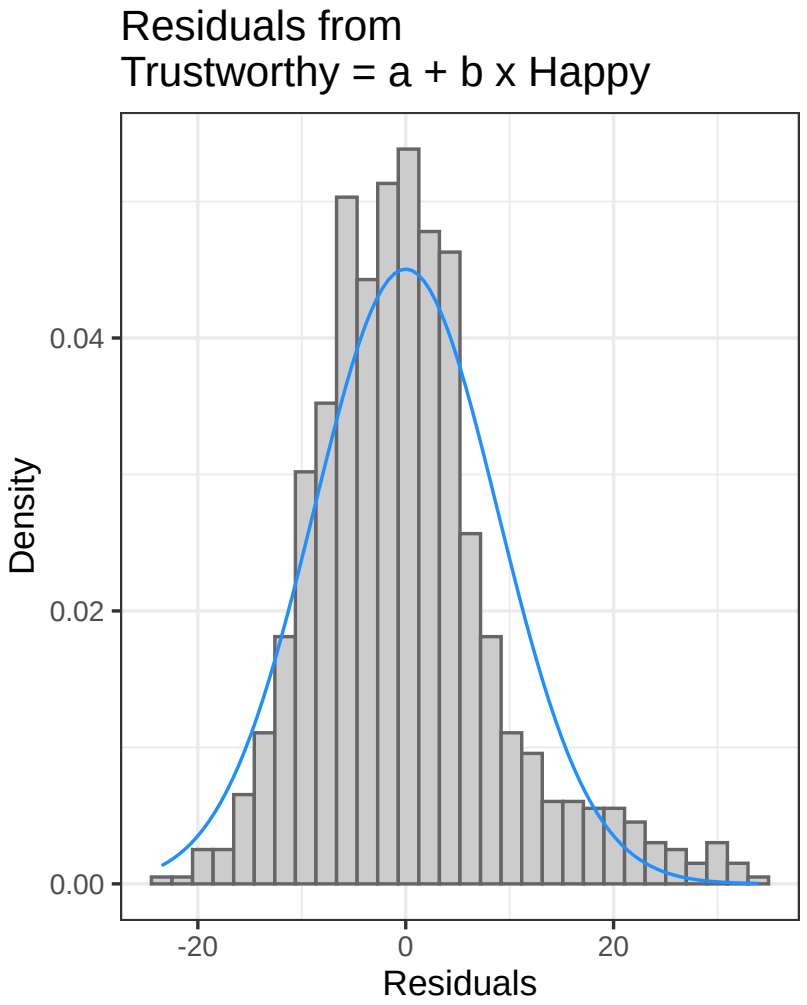


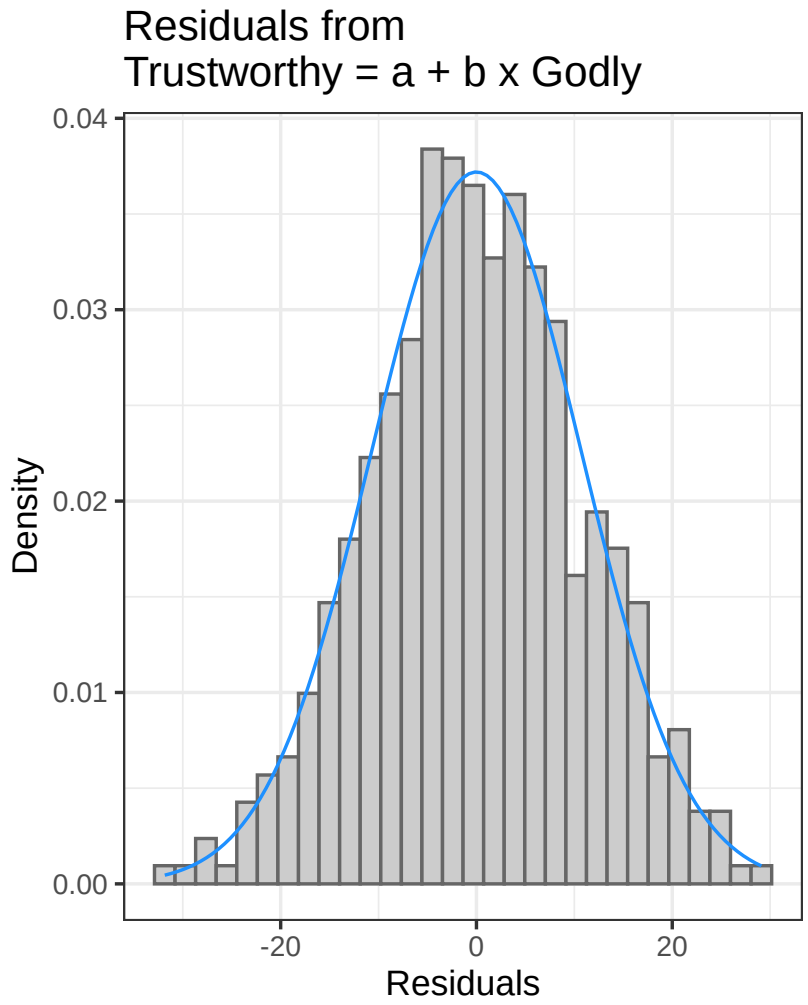
Now we can look at the residuals as a whole.

Normality


Residuals from Trustworthy = a + b x Happy
Density
0.04
0.02
0.00
-20
0
20
Residuals

Normality


Residuals from
Trustworthy = a + b x Happy
Density
0.04
0.02
0.00
-20
0
20
Residuals


Residuals from
Trustworthy = a + b x Godly
Density
0.04
0.03
0.02
0.01
0.00
-20
0
20
Residuals

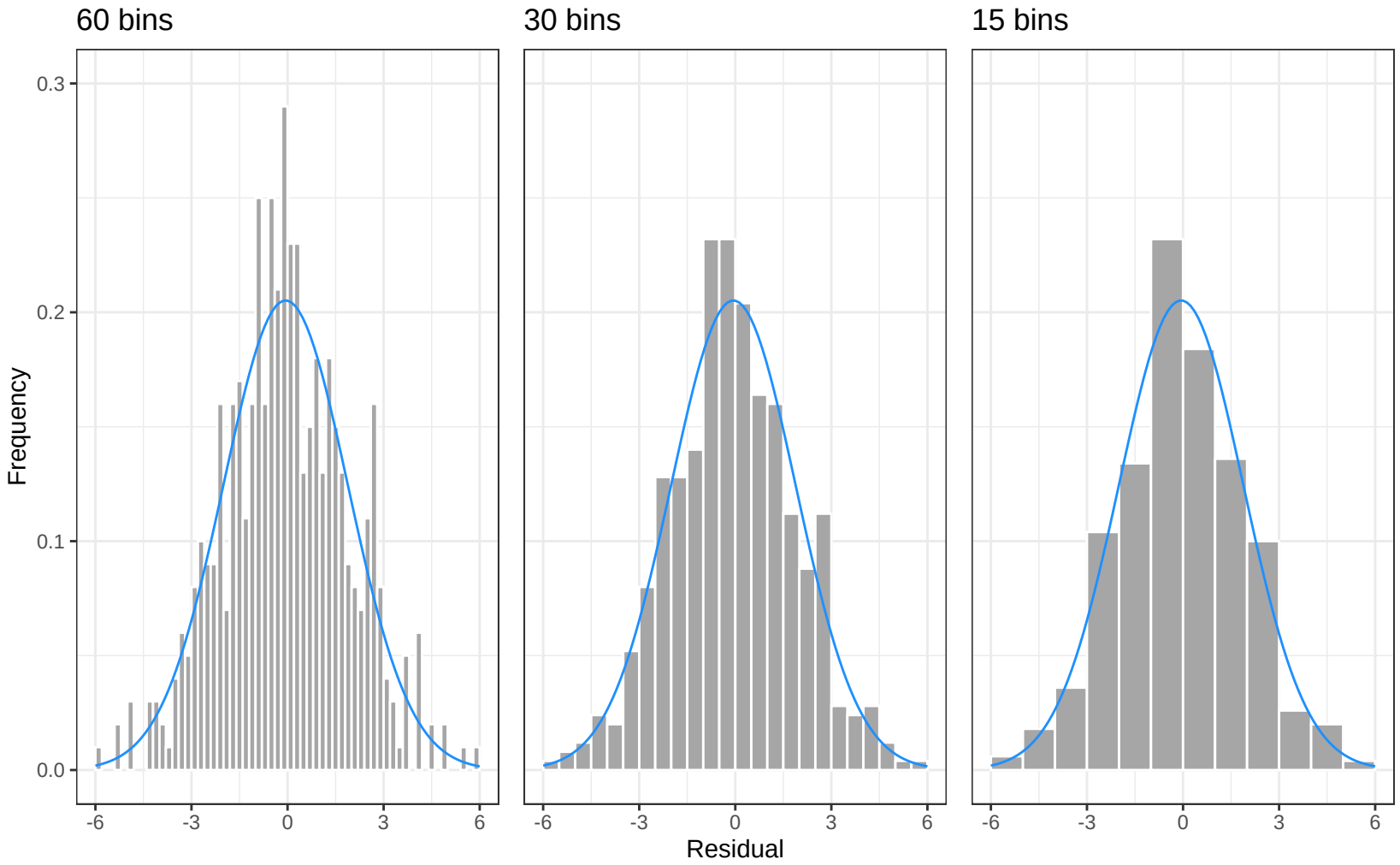
Residual histograms

The histogram of (the residuals of) Godly looks “more normal” than that of Happy.

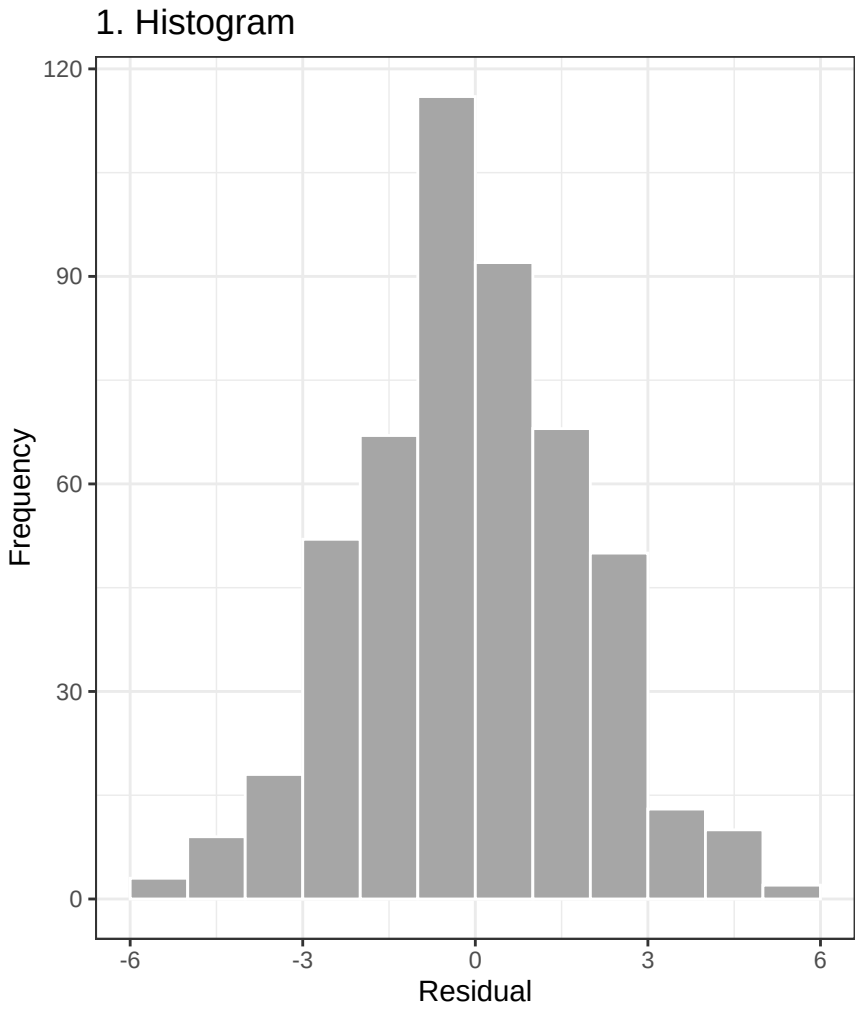
- Godly: fairly symmetric and bell-shaped.
- Happy: a longer right tail.

While histograms are useful, but their appearance depends on the bins.

Histograms depend on the bins

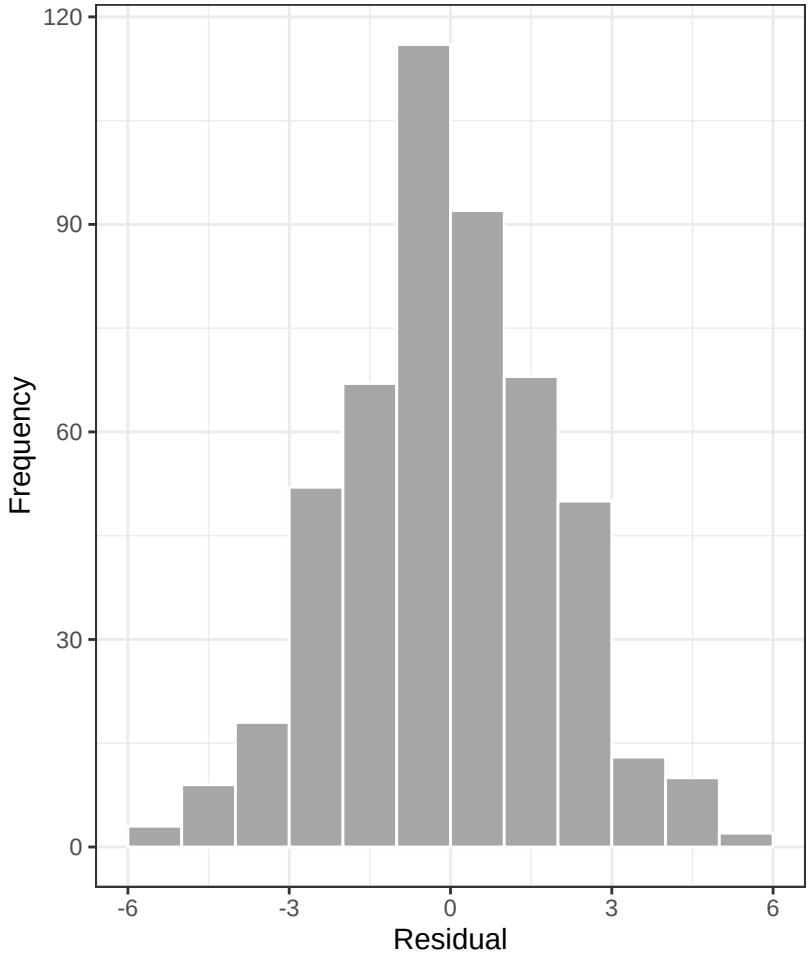


From histograms to QQ plots

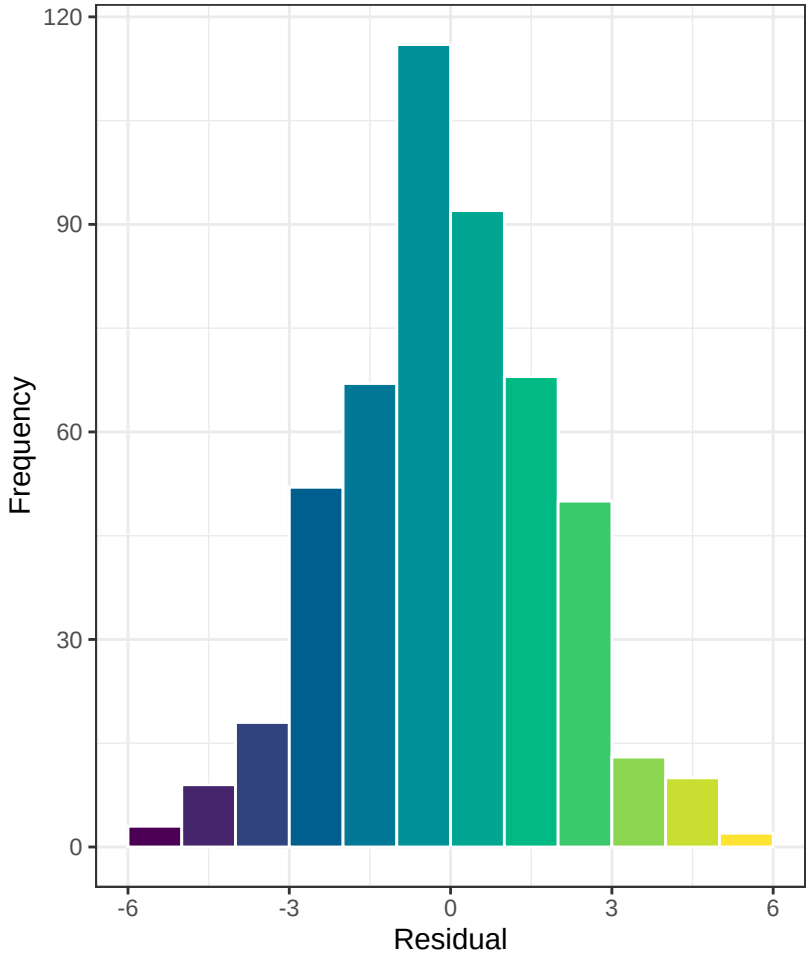


From histograms to QQ plots

1. Histogram



2. From left to right



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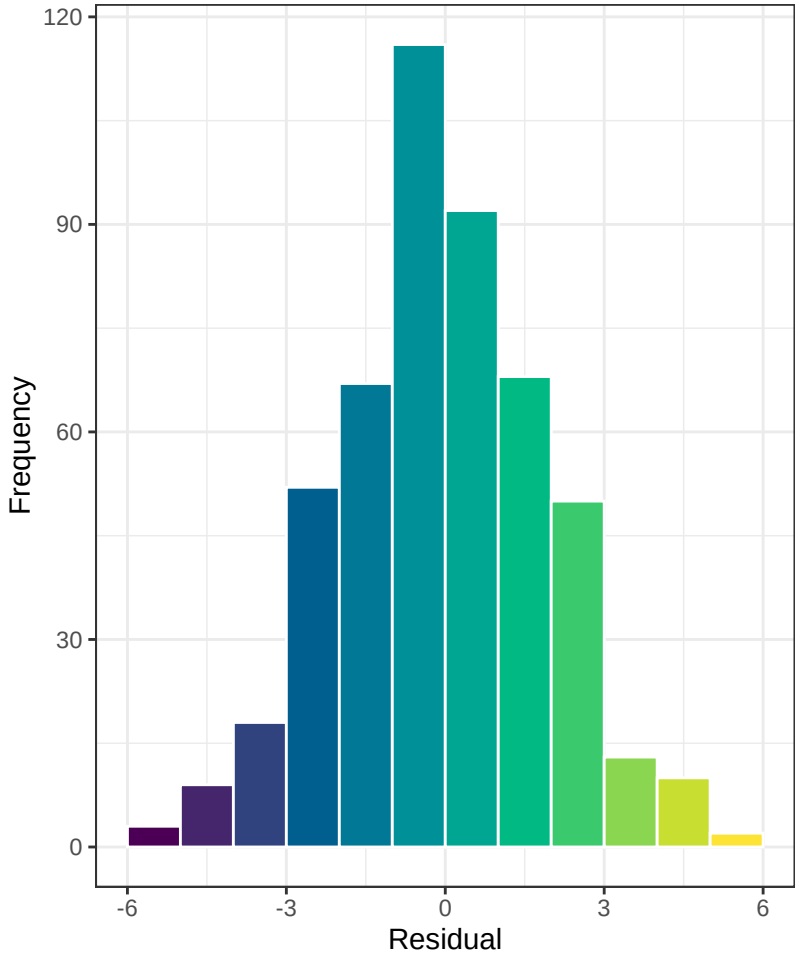
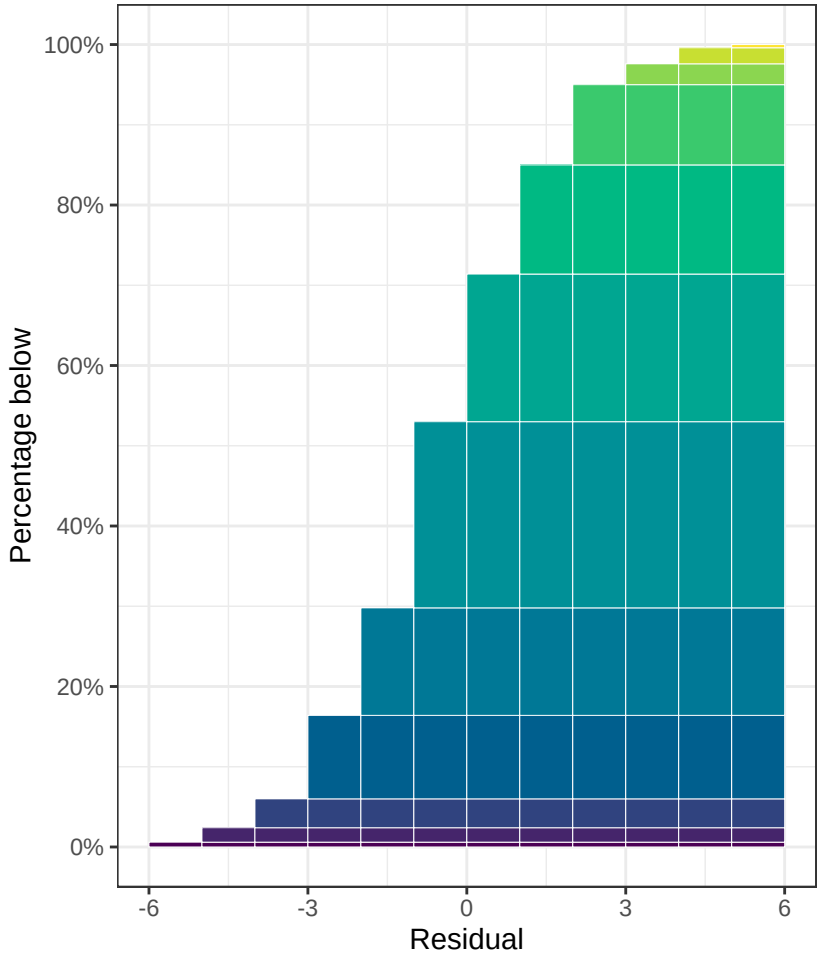


3. Cumulative distribution



From histograms to QQ plots

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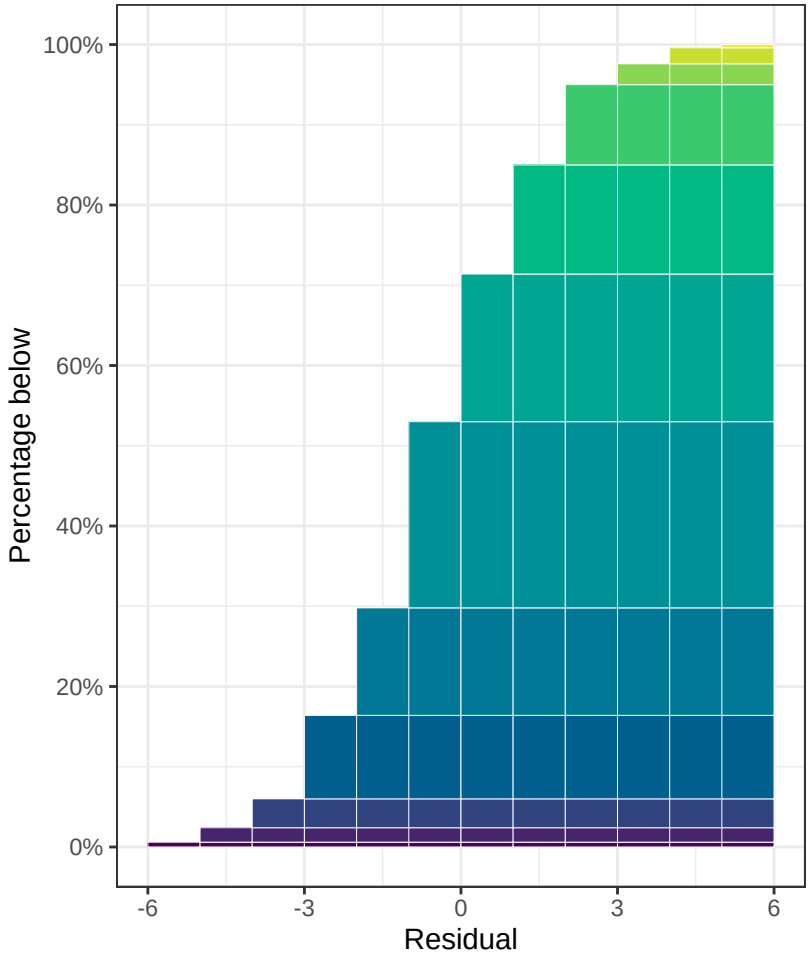
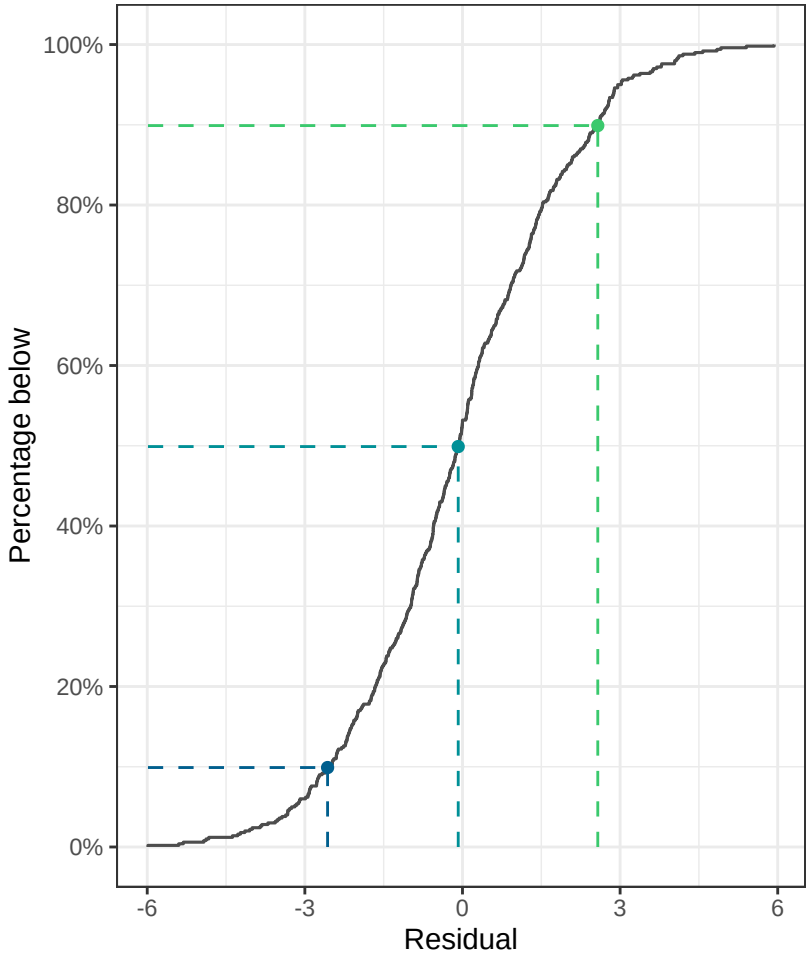
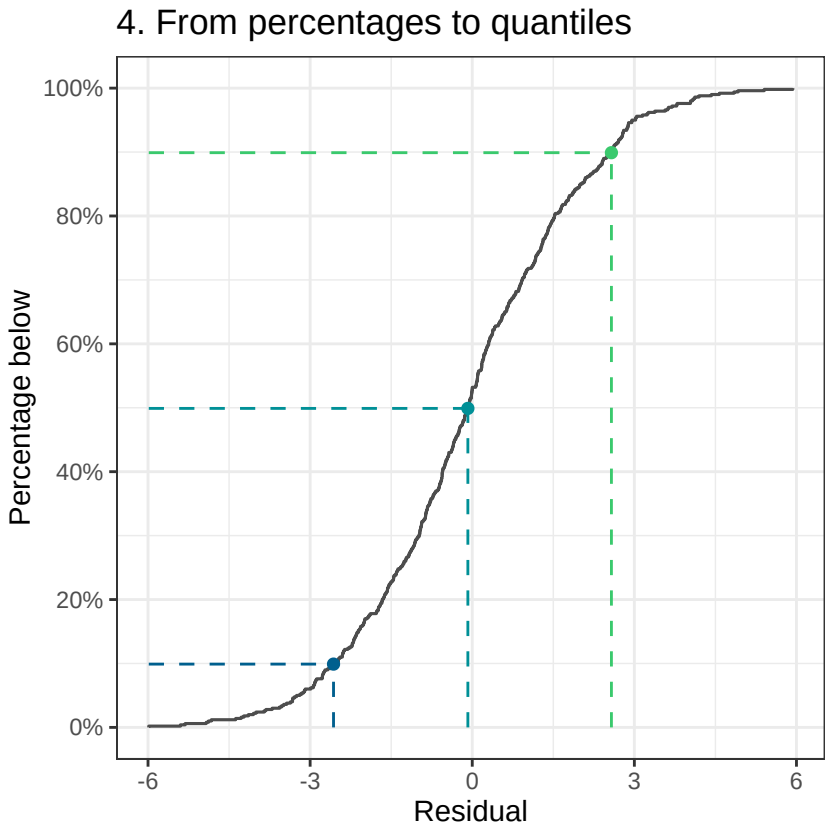
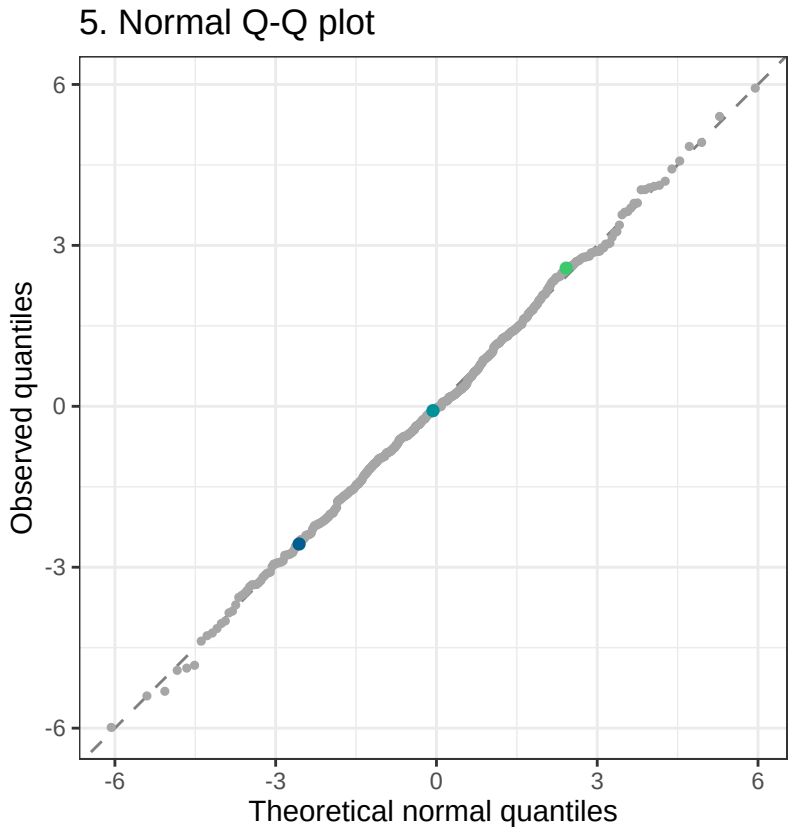


4. From percentages to quantiles



From histograms to QQ plots





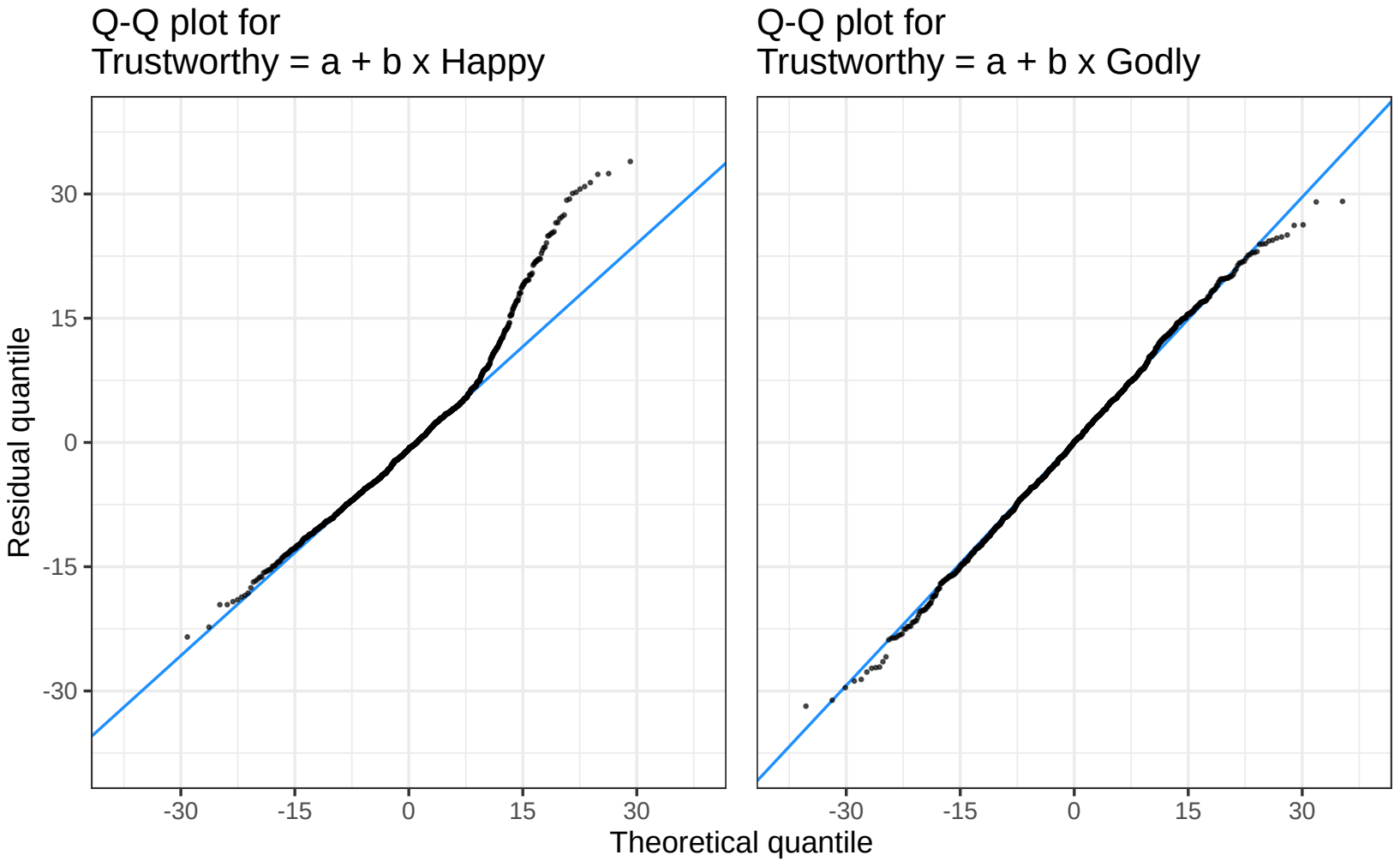
How to read a normal Q-Q plot

A Q-Q plot compares:

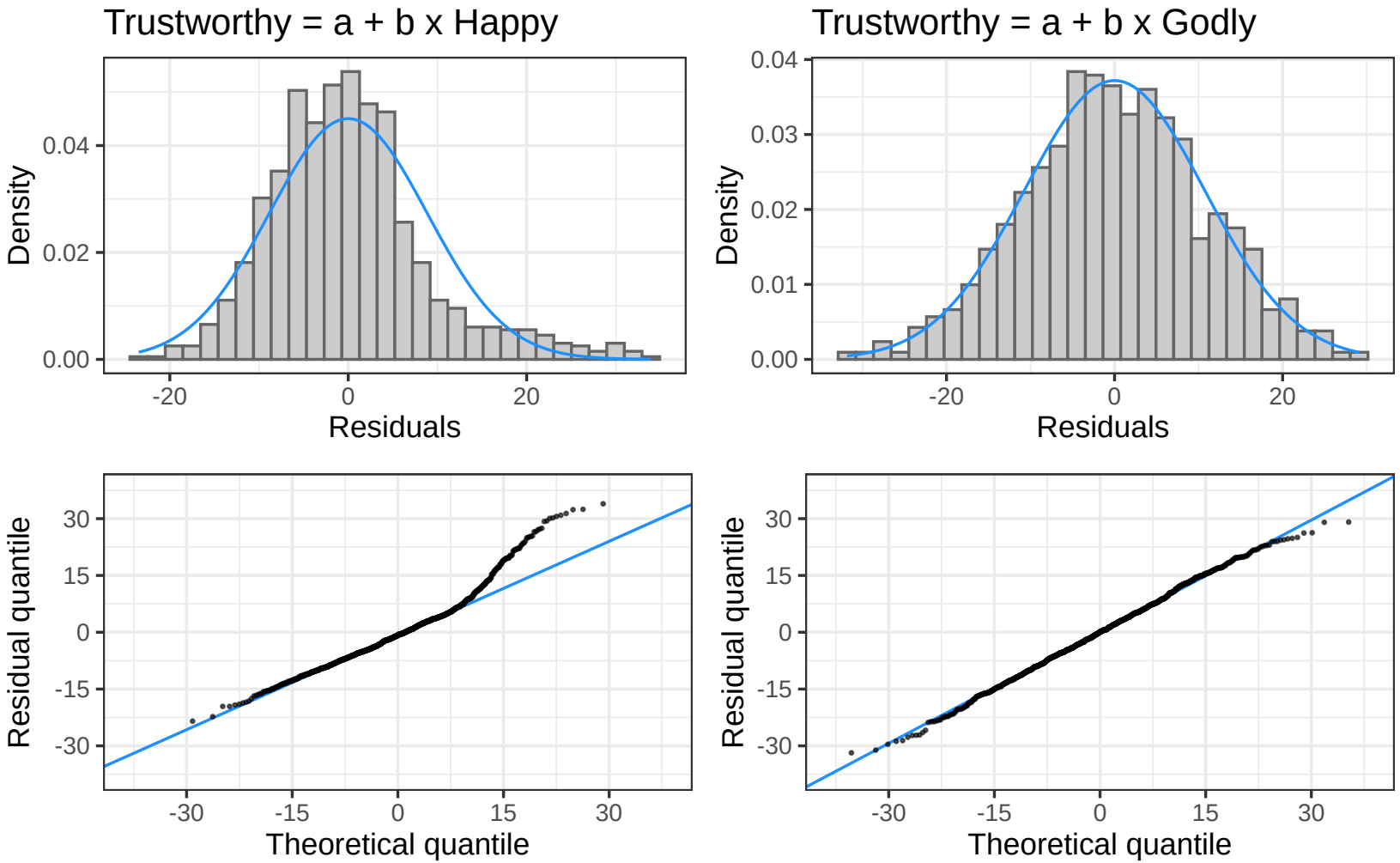
- The percentiles of our residuals.
- The percentiles we would expect from a normal distribution.

- Approximately normal: points lie on a straight line.
- Skewness or unusual tails: systematic departures from that line.
- Some deviation is expected in a sample.

QQ-plots for the Faces



Histograms and QQ-plots



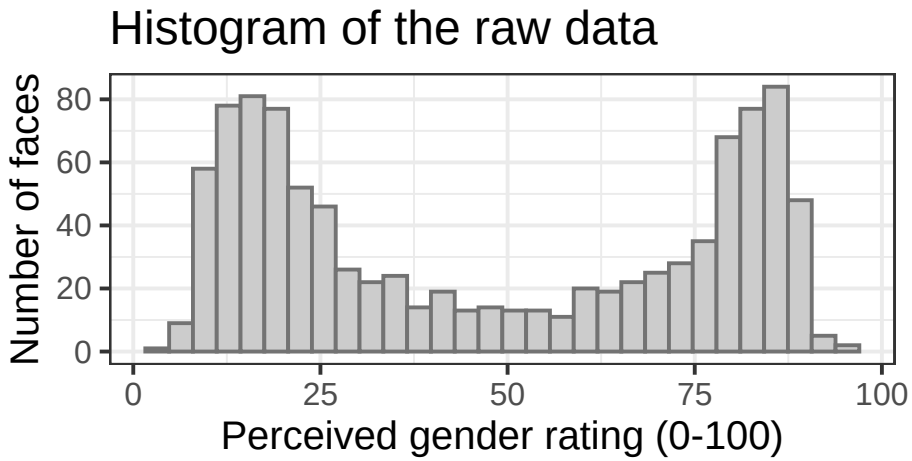
Faces: the predictor need not be normal

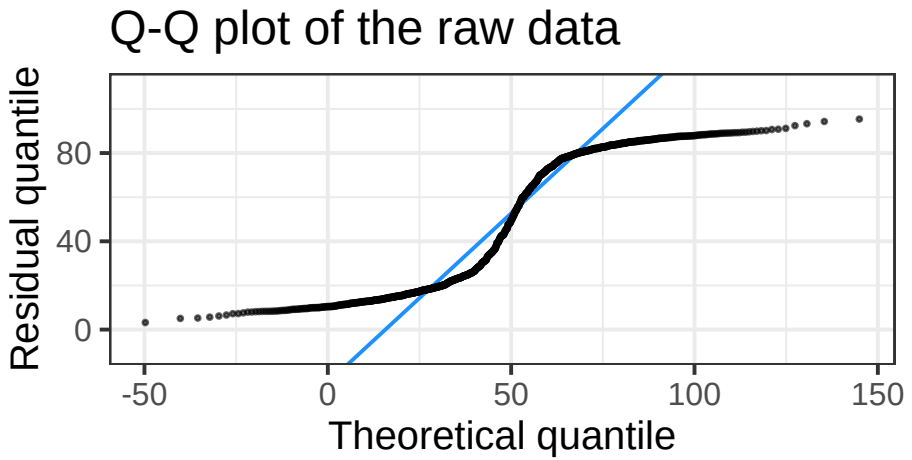
Let's consider the regression

$$\widehat{\text{trustworthy}} = a + b \times \text{gender}$$

- gender was scored on a scale from 0 to 100.
  - Like all other variables.

Gender clearly has two peaks and is absolutely not normal.





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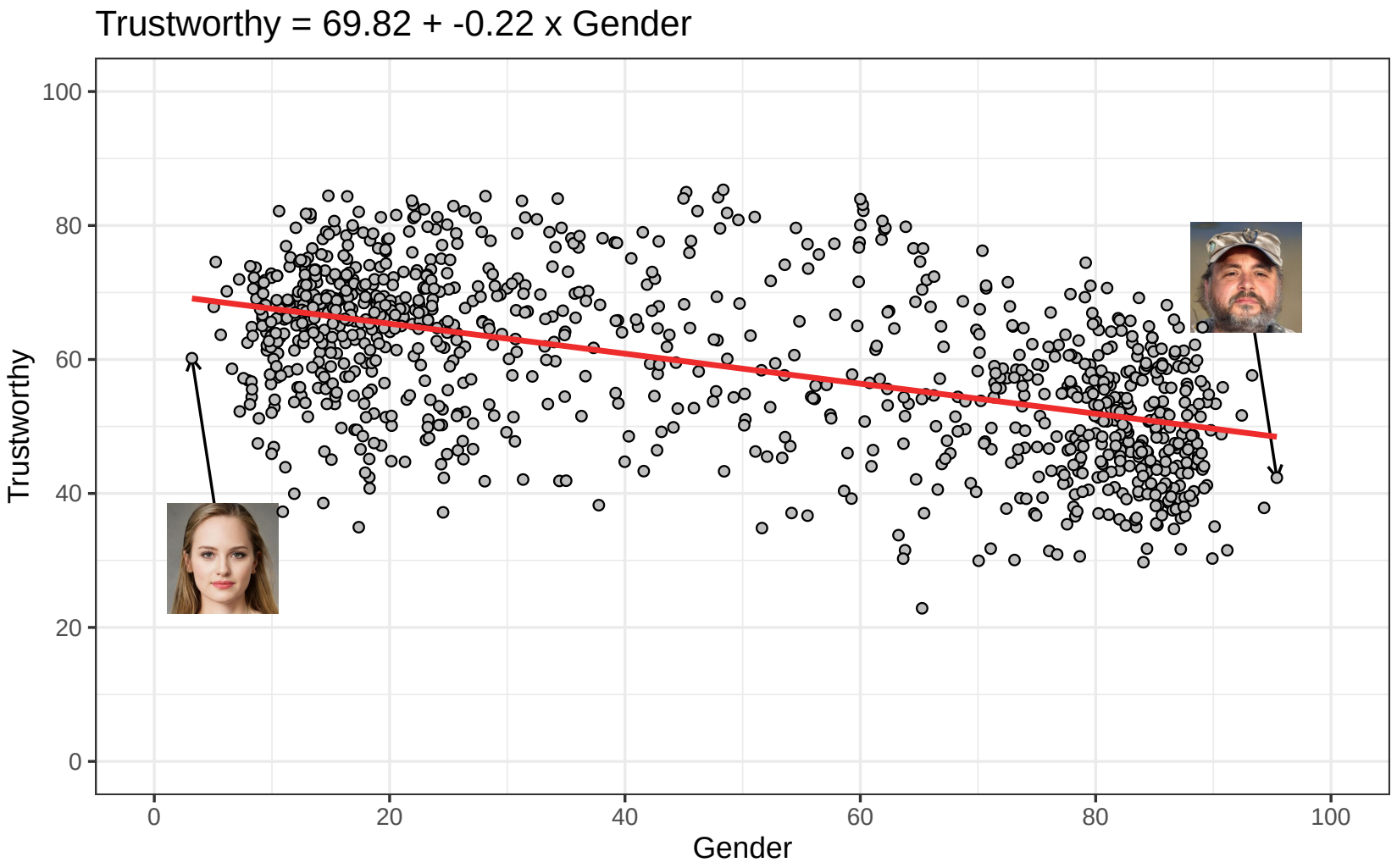

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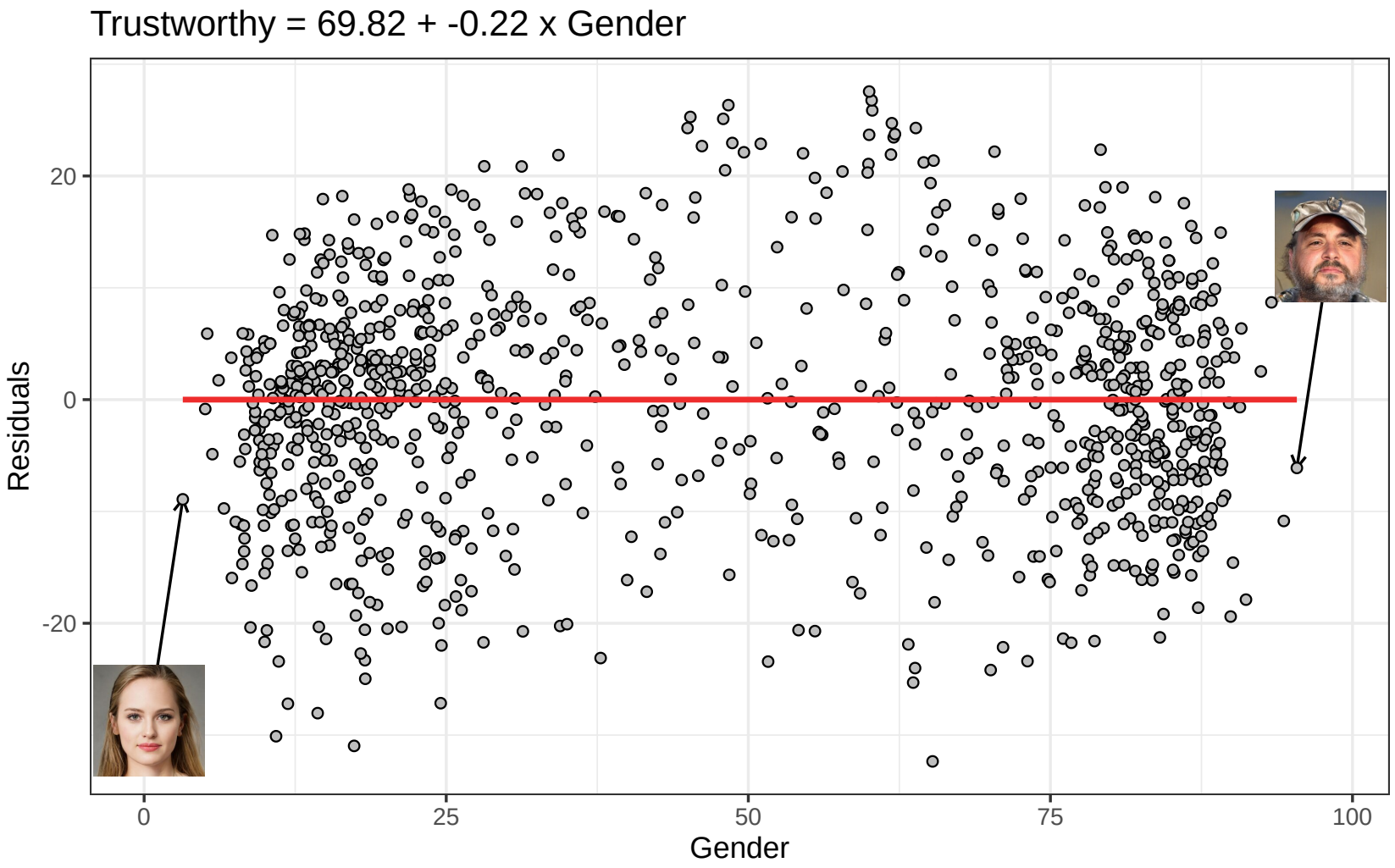

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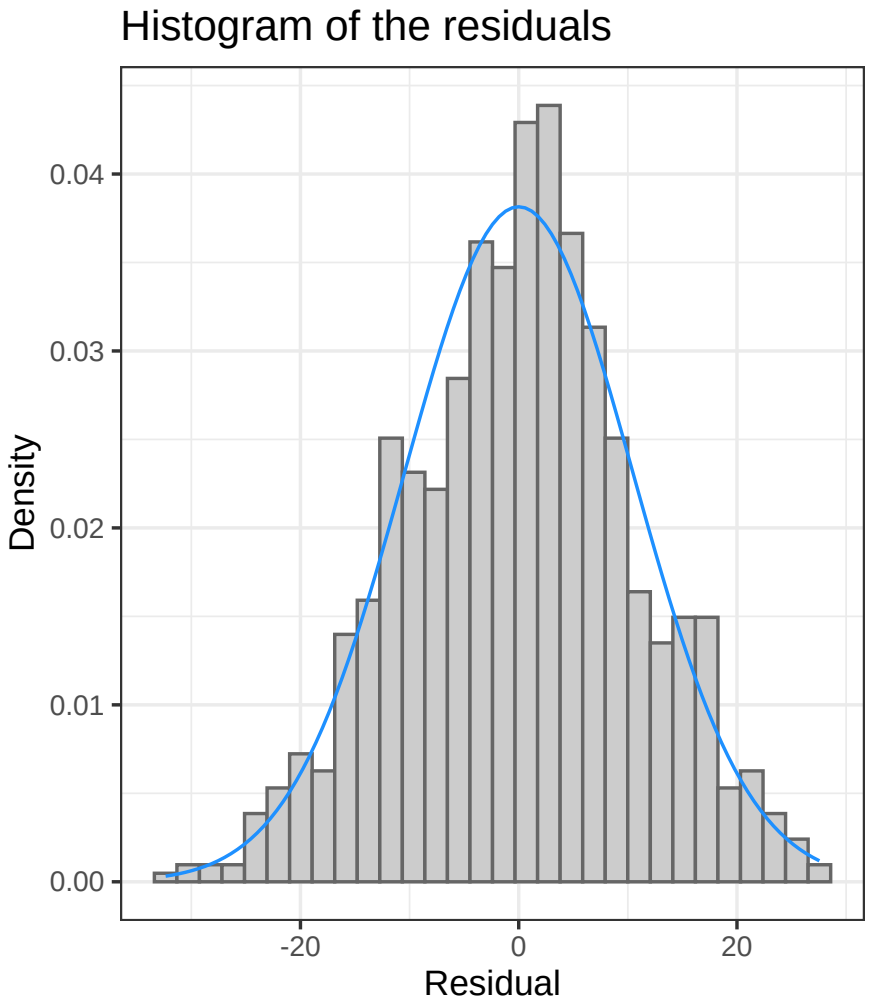

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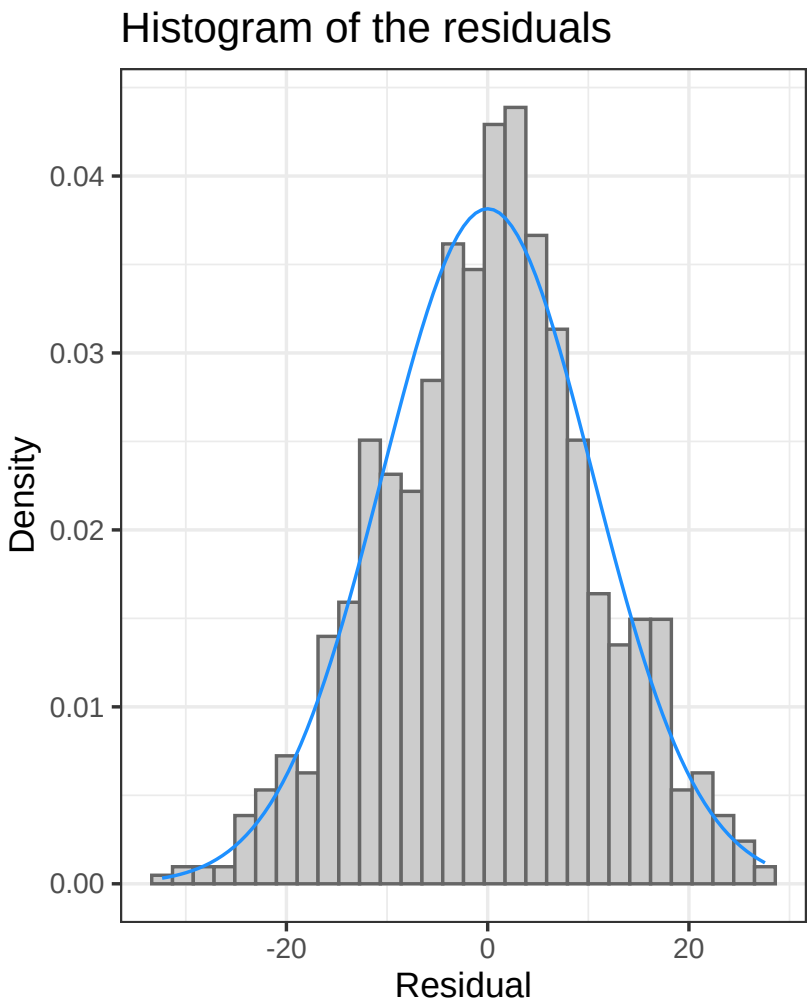

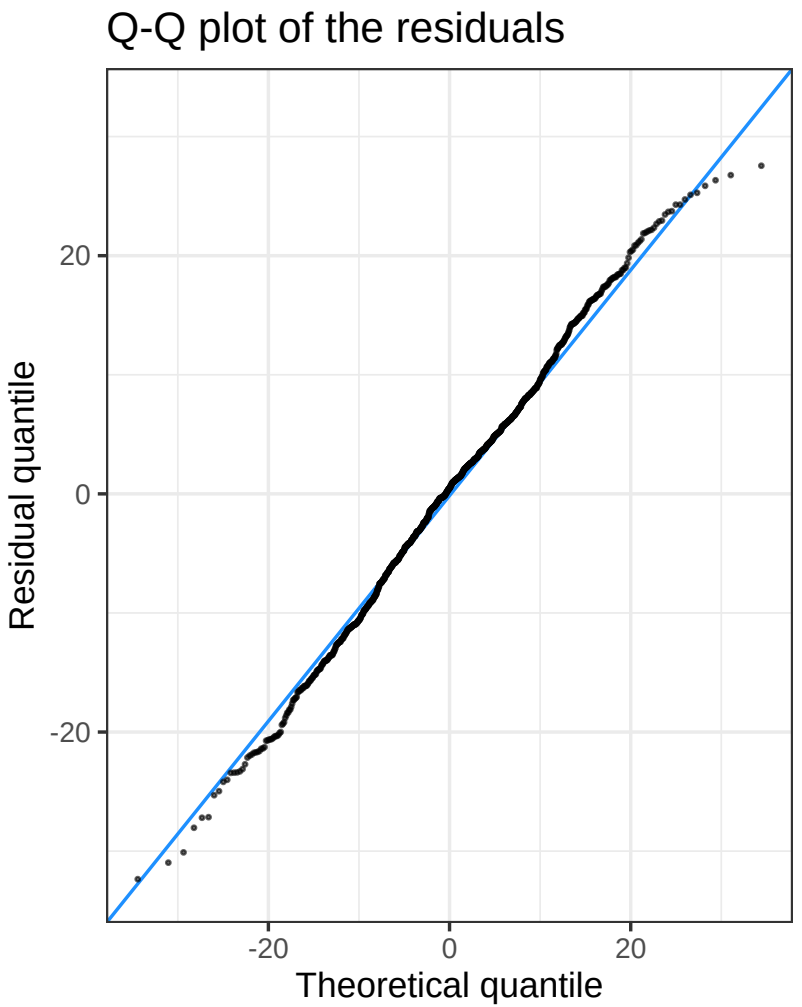

Faces: the predictor need not be normal

- The independent variable Gender is absolutely not normally distributed
- But the residuals look fine in the Q-Q plot

Today

Recap
Assumptions
Residuals
Normality
Constant Error Variance
Independence
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Homoscedasticity

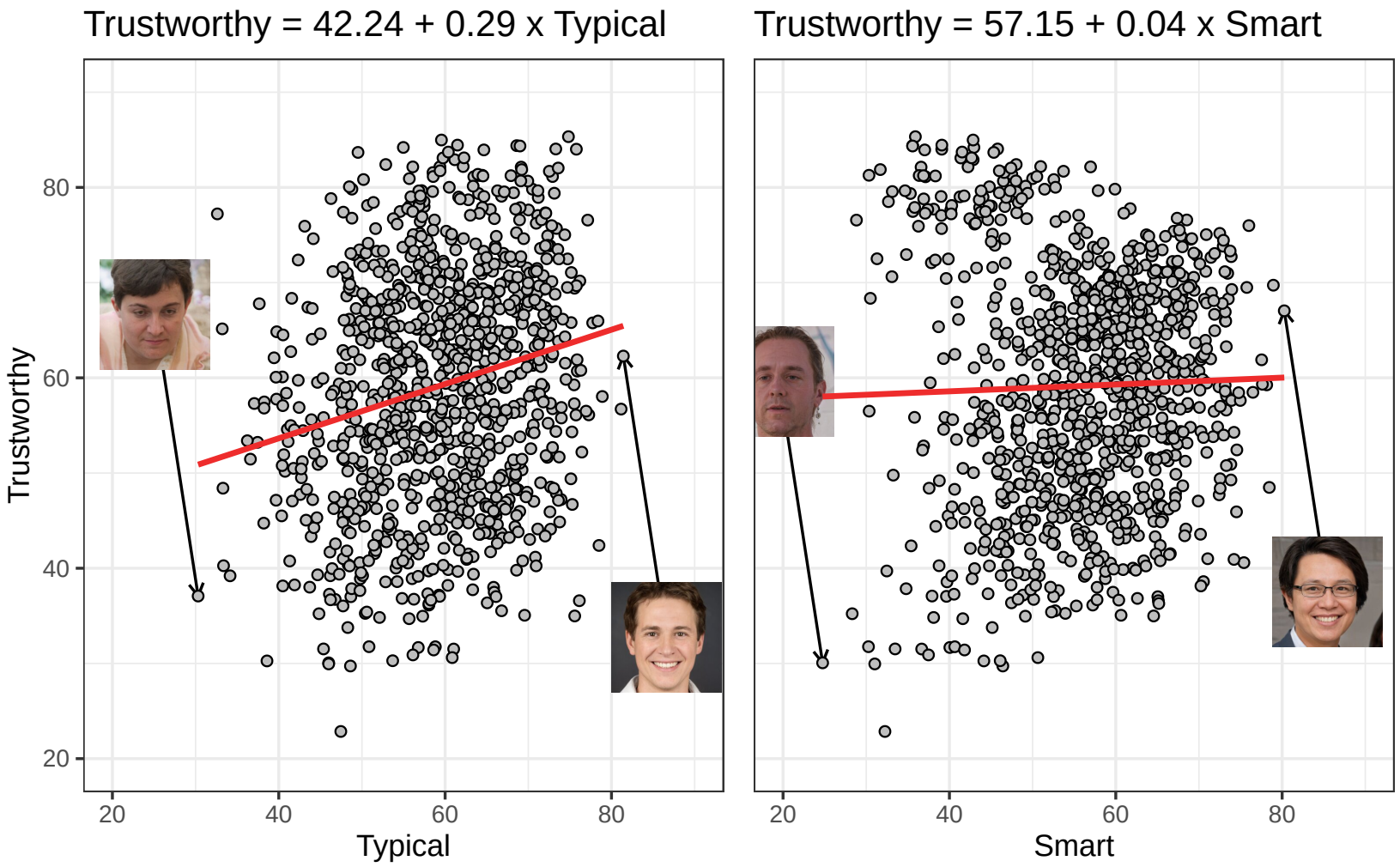
The model uses **one error variance** for all predictor values.

At every x_i, we expect a similar spread of the residuals.

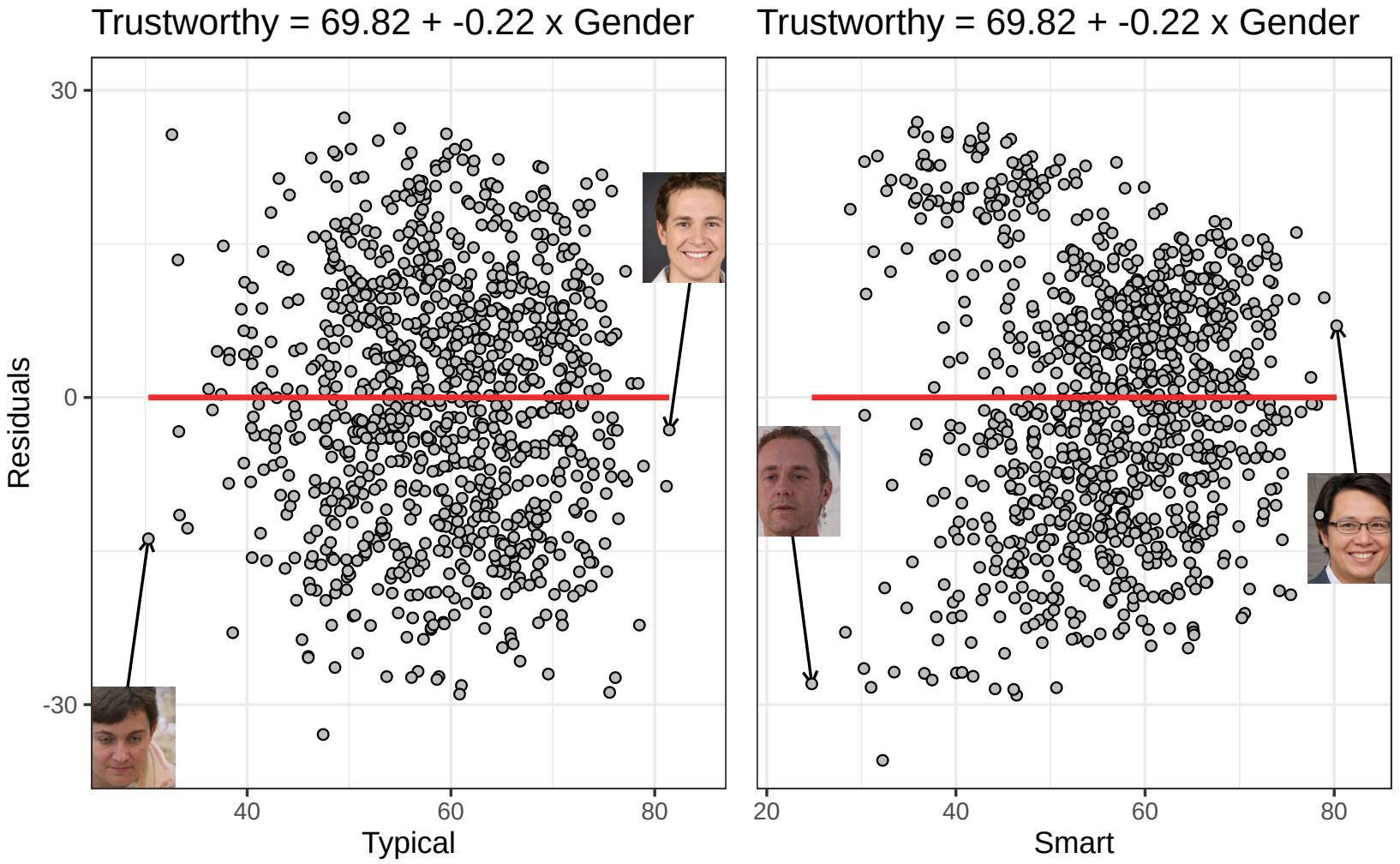
- Homoscedasticity: the same error variance across x.
- Heteroscedasticity: error variance changes across x.

The residuals can look normal overall, but still heteroscedastic.

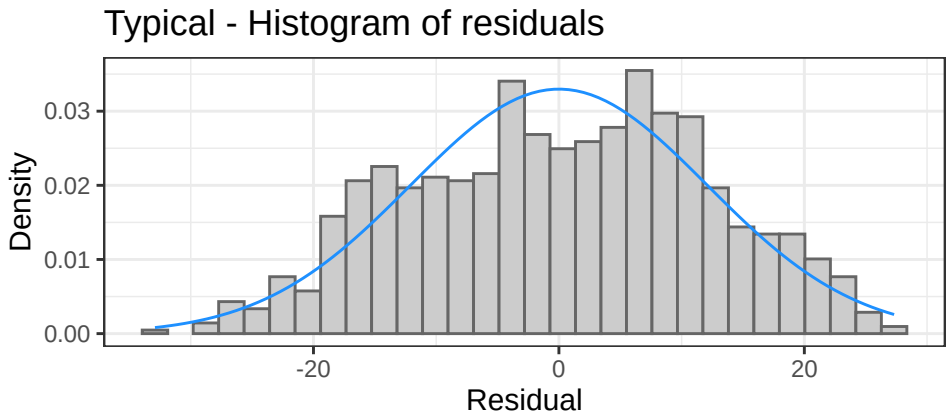
Typical and Smart - Regression

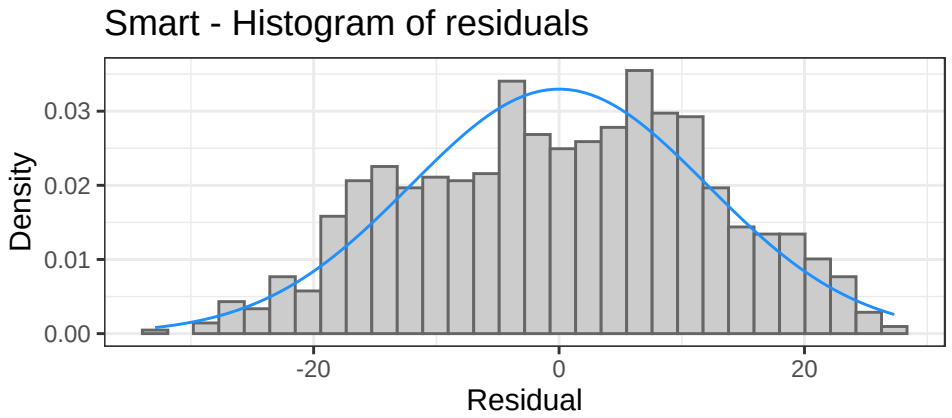


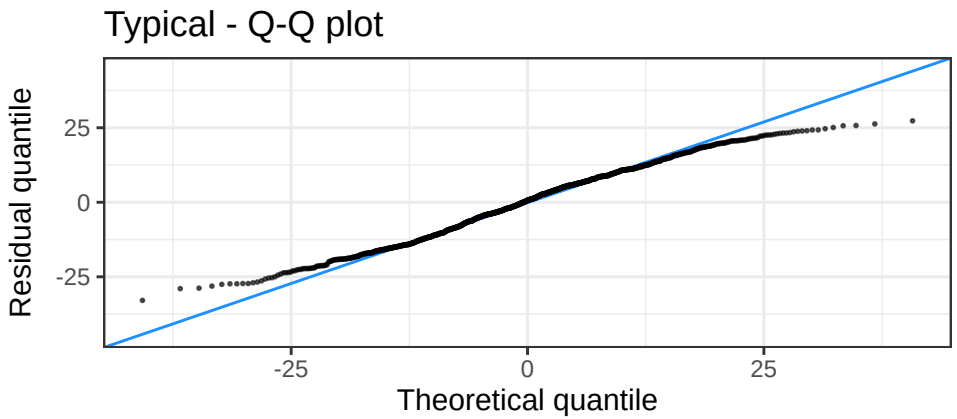
Typical and Smart - Residuals

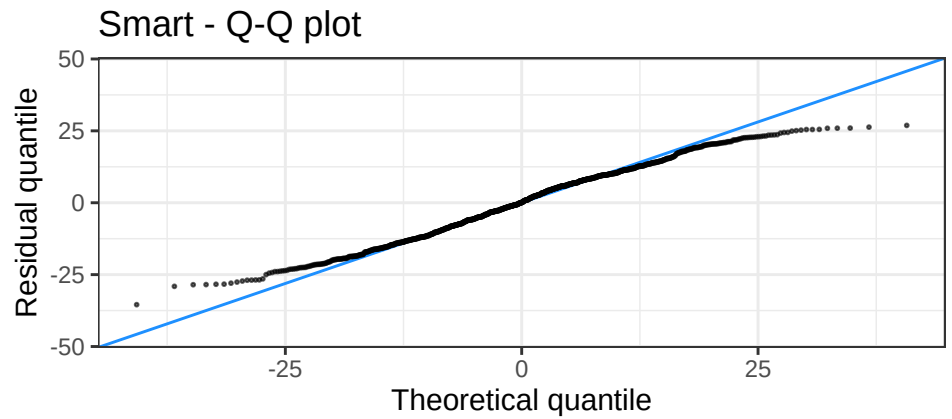


Typical and Smart - Normality

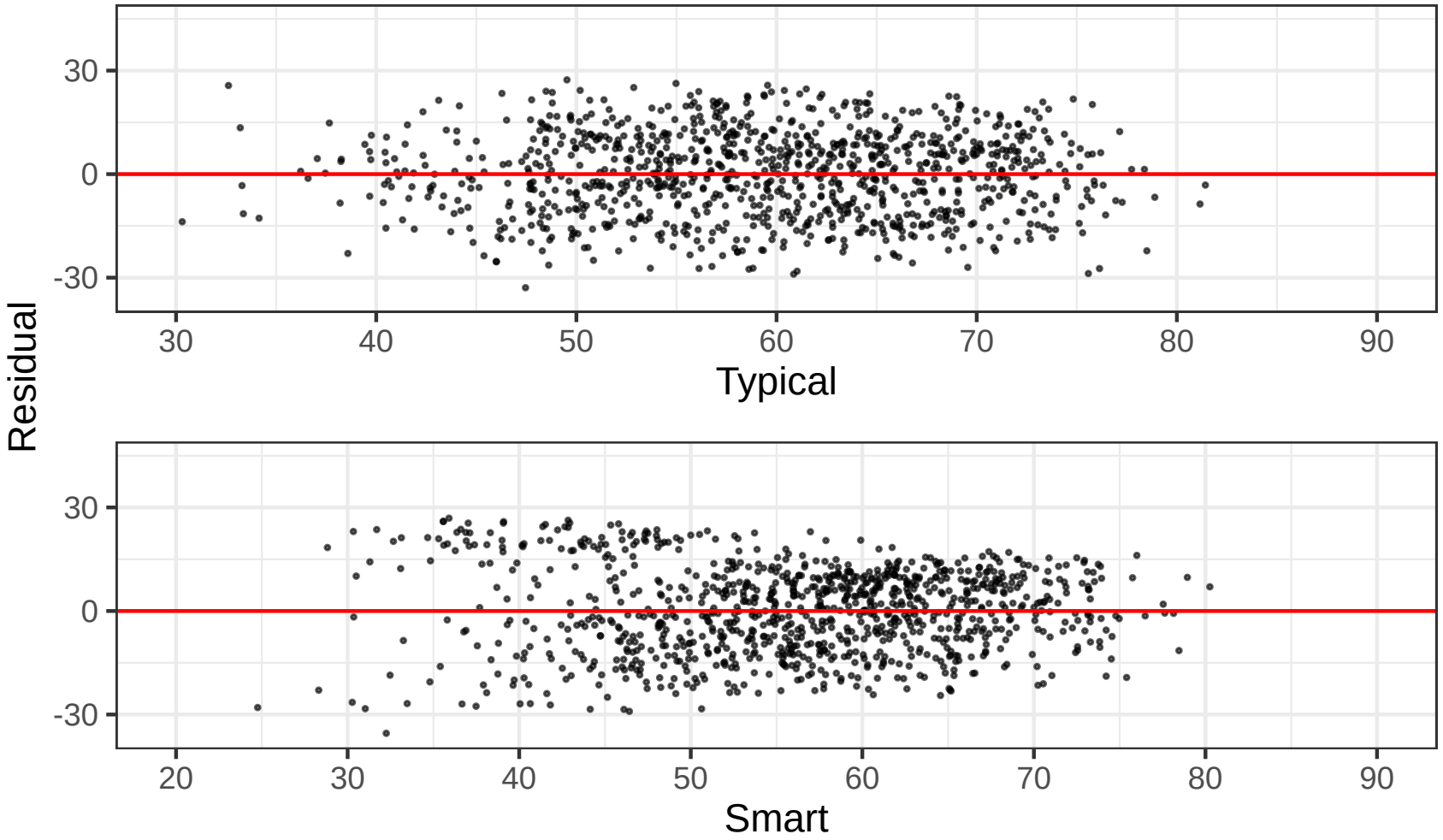








Examine the residuals' spread



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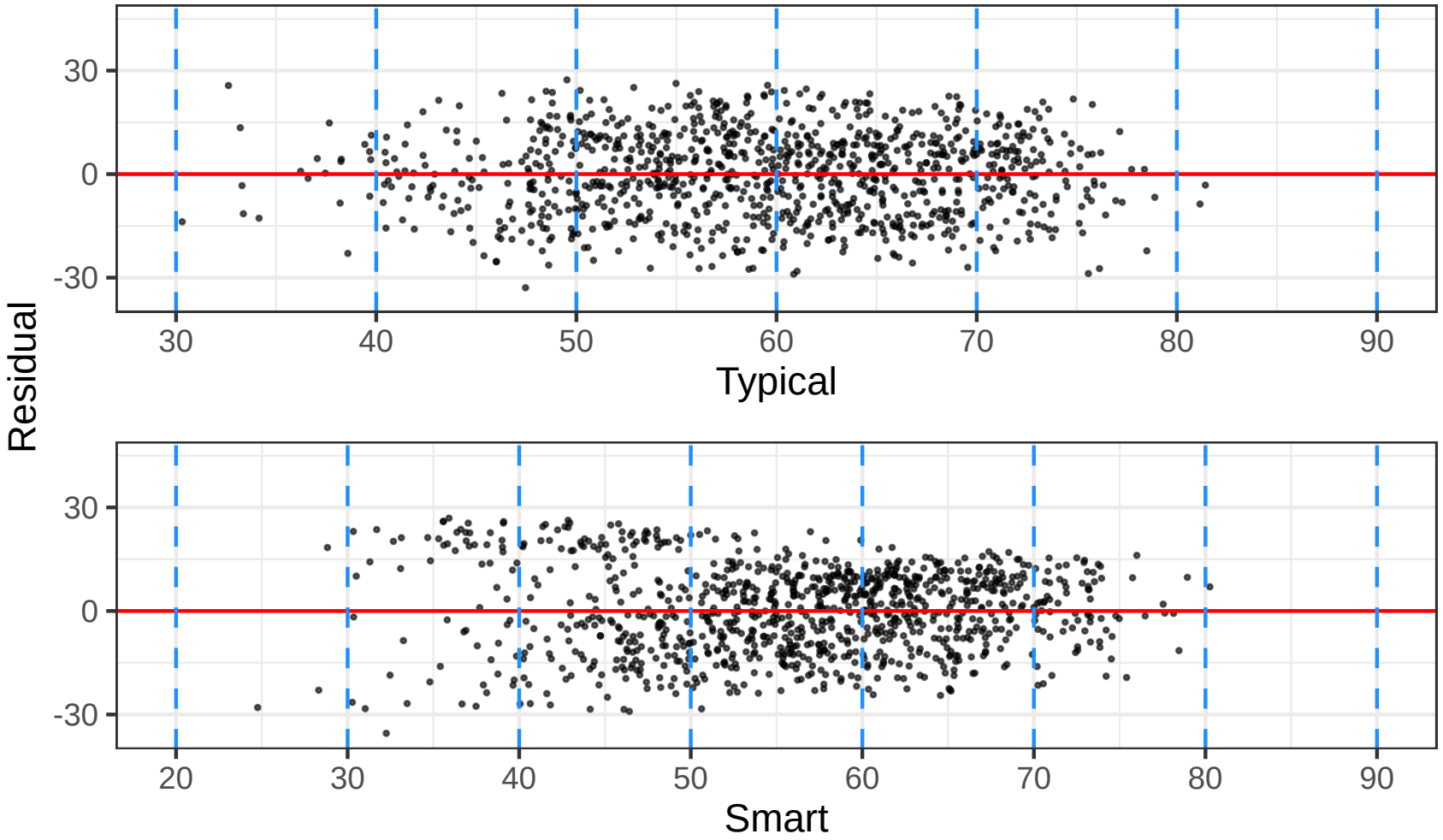

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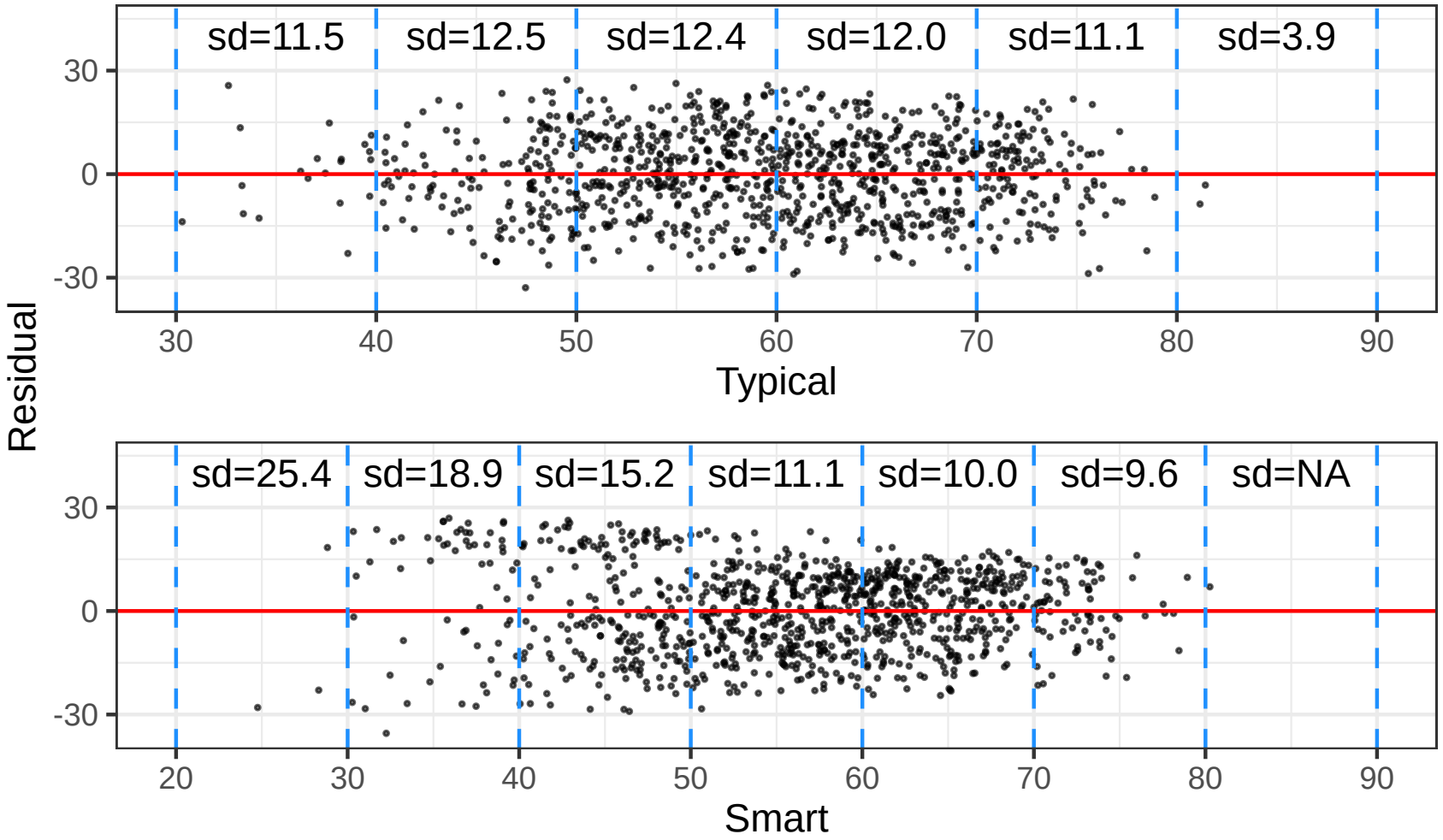


Examine the residuals' spread

- Typical: spread is relatively similar across much of the predictor range.
- Smart: residuals tend to be more spread out at lower ratings.

In general, the residuals can widen or narrow. Both suggest unequal error variance.

Why does it matter?

- The usual standard errors assume constant error variance.
- If this assumption is violated, significance tests may be misleading.

Zooming out

We’ve examined three assumptions using plots.

But can we also use statistical tests?

Well, for normality we can use the Shapiro–Wilk test!

Shapiro-Wilk: testing normality

$$\mathcal{H}_0 : \text{the distribution is normal}$$
$$\mathcal{H}_1 : \text{the distribution is not normal}$$

SPSS gives you a test-statistic W and a p-value.

- If $p < .05$, there is evidence against normality.

You know by now:

- what a test statistic is
- what a p-value is
- and therefore how to interpret the Shapiro Wilk test.

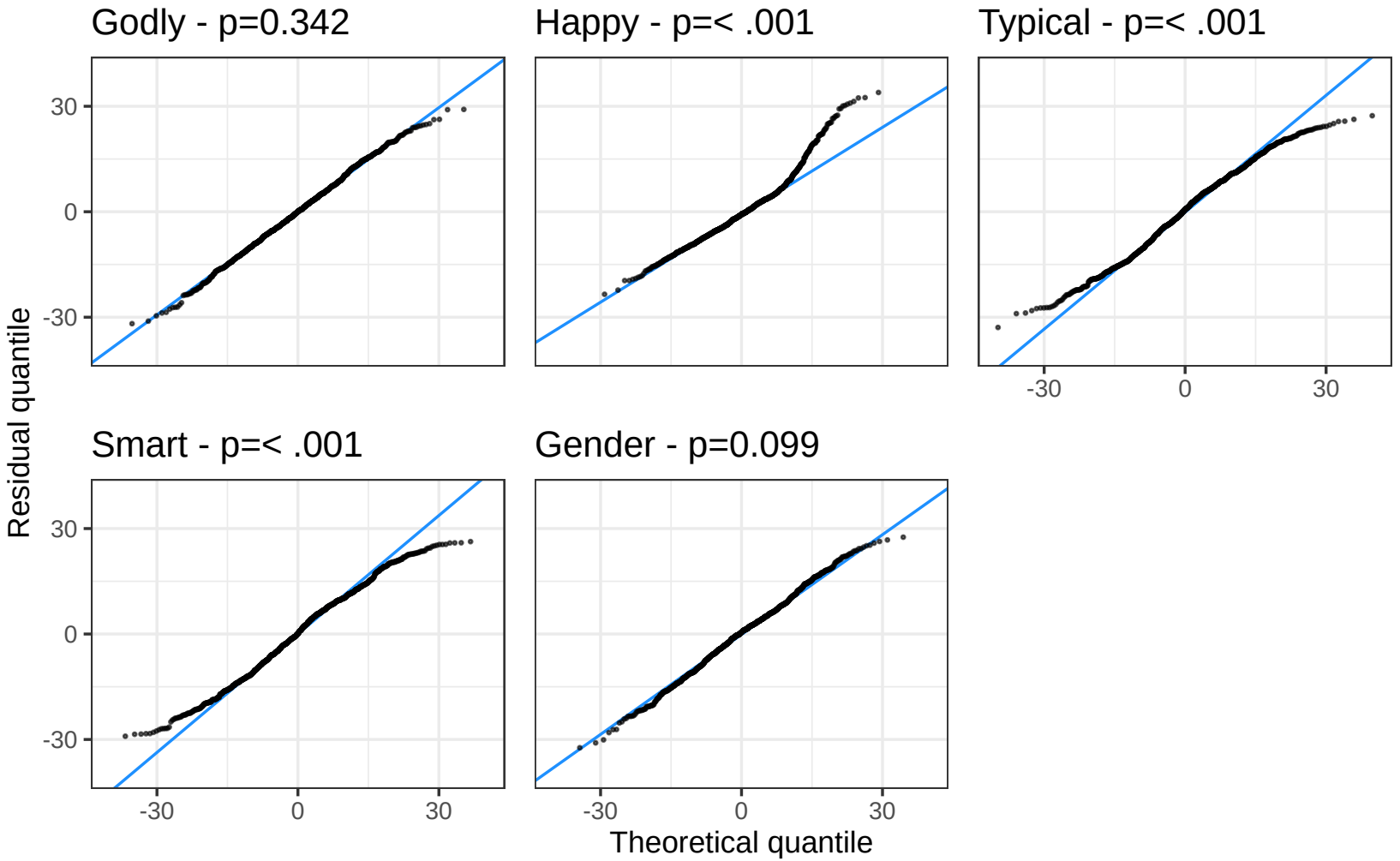
Faces: Shapiro-Wilk results

Predictor	N	Shapiro-Wilk W	p
godly	1004	0.998	0.342
happy	1004	0.956	< .001
typical	1004	0.988	< .001
smart	1004	0.989	< .001
gender	1004	0.997	0.099

Each row is a test on the *residuals* of the regression predicting trustworthiness with that variable.

Let's compare these results with the Q-Q plots.

Faces: Shapiro-Wilk vs. Q-Q plots



Sample size changes what a test detects

- Small sample: an important departure can go undetected.
- Large sample: a minor departure can be statistically significant.

Often, the Q-Q plots can be more informative than the statistical tests.

In the faces data, with $N = 1,004$, many shapiro wilk tests are significant.

But many Q-Q plots look fine.

There is a parallel with the “significant” $R = 0.01$ here.

Just because an assumption is violated according to a significance test, does not mean it actually is a problem.

The type and severity of the violation matter.

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Independence

$$Y_i = \alpha + \beta \times x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\epsilon^2)$$

Independence: Prediction errors of different observations are unrelated.

What does one row represent?

Independent observations contribute independent information.

Study	Potential dependence
One measurement from each independently sampled person	No dependence built into the design
Two measurements from each person	Measurements share a person
Students from the same classroom	Students share a classroom

- This is all about how the data were collected.
- We usually cannot assess this using statistics, but have to reason about the design

Independence in the faces data

Each regression has one row per face, not one row per individual rating.

We could wonder:

- Are the same or closely related face stimuli included more than once?
- Did participants rate multiple faces, potentially introducing shared rating tendencies?
- How were ratings combined into the face averages?

Just because we average many ratings does not mean the ratings of the faces become independent.

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We've learned

Assumption	How do we check it?
Linearity	Look for patterns in the residual plot
Outliers	Cook's distance
Normality	Examine the Q-Q plot
Homoscedasticity	Examine the spread of residuals
Independence	Consider how the data were collected

Can we prove an assumption?

No.

Assumptions concern the population. We examine sample diagnostics.

- A plot can reveal important departures.
- A formal test can provide evidence against an assumption.
- Neither can prove that an assumption holds exactly.

In large samples, the degree of the violations matters much more than a significance test.

Large sample sizes mean small deviations are quickly significant

When assumptions are violated

Ideal world: fit a different model that doesn’t make the assumption

This course: report it properly.

Next weeks

- Independent samples t-test (next week)
  - Here we do have a solution when homoscedasticity is violated
- ANOVA (both weeks after that)